desired frequency response $H_{dr}(\omega)$ and the weighting function $W(\omega)$ are piecewise constant, we have

$$\frac{dE(\omega)}{d\omega} = \frac{d}{d\omega}\{W(\omega)[H_{dr}(\omega) - H_r(\omega)]\}$$
$$= -\frac{dH_r(\omega)}{d\omega} = 0$$

Consequently, the frequencies $\{\omega_i\}$ corresponding to the peaks of $E(\omega)$ also correspond to peaks at which $H_r(\omega)$ meets the error tolerance. Since $H_r(\omega)$ is a trigonometric polynomial of degree L, for Case 1, for example,

$$\begin{aligned} H_r(\omega) &= \sum_{k=0}^{L} \alpha(k)\cos \omega k \\ &= \sum_{k=0}^{L} \alpha(k)\left[\sum_{n=0}^{k} \beta_{nk}(\cos\omega)^n\right] \\ &= \sum_{k=0}^{L} \alpha'(k)(\cos\omega)^k \end{aligned} \tag{8.2.71}$$

it follows that $H_r(\omega)$ can have at most $L-1$ local maxima and minima in the open interval $0 < \omega < \pi$. In addition, $\omega = 0$ and $\omega = \pi$ are usually extrema of $H_r(\omega)$ and, also, of $E(\omega)$. Therefore, $H_r(\omega)$ has at most $L+1$ extremal frequencies. Furthermore, the band-edge frequencies ω_p and ω_s are also extrema of $E(\omega)$, since $|E(\omega)|$ is maximum at $\omega = \omega_p$ and $\omega = \omega_s$. As a consequence, there are at most $L+3$ extremal frequencies in $E(\omega)$ for the unique, best approximation of the ideal lowpass filter. On the other hand, the alternation theorem states that there are at least $L+2$ extremal frequencies in $E(\omega)$. Thus the error function for the lowpass filter design has either $L+3$ or $L+2$ extrema. In general, filter designs that contain more than $L+2$ alternations or ripples are called *extra ripple filters*. When the filter design contains the maximum number of alternations, it is called a *maximal ripple filter*.

The alternation theorem guarantees a unique solution for the Chebyshev optimization problem in (8.2.70). At the desired extremal frequencies $\{\omega_n\}$, we have the set of equations

$$\hat{W}(\omega_n)[\hat{H}_{dr}(\omega_n) - P(\omega_n)] = (-1)^n\delta \qquad n = 0, 1, \ldots, L+1 \tag{8.2.72}$$

where δ represents the maximum value of the error function $E(\omega)$. In fact, if we select $W(\omega)$ as indicated by (8.2.66), it follows that $\delta = \delta_2$.

The set of linear equations in (8.2.72) can be rearranged as

$$P(\omega_n) + \frac{(-1)^n\delta}{\hat{W}(\omega_n)} = \hat{H}_{dr}(\omega_n) \qquad n = 0, 1, \ldots, L+1$$

or, equivalently, in the form

$$\sum_{k=0}^{L} \alpha(k)\cos\omega_n k + \frac{(-1)^n\delta}{\hat{W}(\omega_n)} = \hat{H}_{dr}(\omega_n) \qquad n = 0, 1, \ldots, L+1 \tag{8.2.73}$$

If we treat the $\{a(k)\}$ and δ as the parameters to be determined, (8.2.73) can be expressed in matrix form as

$$\begin{bmatrix} 1 & \cos\omega_0 & \cos 2\omega_0 & \cdots & \cos L\omega_0 & \dfrac{1}{\hat{W}(\omega_0)} \\ 1 & \cos\omega_1 & \cos 2\omega_1 & \cdots & \cos L\omega_1 & \dfrac{-1}{\hat{W}(\omega_1)} \\ & \vdots & & & & \\ 1 & \cos\omega_{L+1} & \cos 2\omega_{L+1} & \cdots & \cos L\omega_{L+1} & \dfrac{(-1)^{L+1}}{\hat{W}(\omega_{L+1})} \end{bmatrix} \begin{bmatrix} \alpha(0) \\ \alpha(1) \\ \vdots \\ \alpha(L) \\ \delta \end{bmatrix} = \begin{bmatrix} \hat{H}_{dr}(\omega_0) \\ \hat{H}_{dr}(\omega_1) \\ \vdots \\ \hat{H}_{dr}(\omega_{L+1}) \end{bmatrix} \tag{8.2.74}$$

Initially, we know neither the set of extremal frequencies $\{\omega_n\}$ nor the parameters $\{\alpha(k)\}$ and δ. To solve for the parameters, we use an iterative algorithm, called the *Remez exchange algorithm* [see Rabiner et al. (1975)], in which we begin by guessing at the set of extremal frequencies, determine $P(\omega)$ and δ, and then compute the error function $E(\omega)$. From $E(\omega)$ we determine another set of $L+2$ extremal frequencies and repeat the process iteratively until it converges to the optimal set of extremal frequencies. Although the matrix equation in (8.2.74) can be used in the iterative procedure, matrix inversion is time consuming and inefficient.

A more efficient procedure, suggested in the paper by Rabiner et al. (1975), is to compute δ analytically, according to the formula

$$\delta = \frac{\gamma_0 \hat{H}_{dr}(\omega_0) + \gamma_1 \hat{H}_{dr}(\omega_1) + \cdots + \gamma_{L+1}\hat{H}_{dr}(\omega_{L+1})}{\dfrac{\gamma_0}{\hat{W}(\omega_0)} - \dfrac{\gamma_1}{\hat{W}(\omega_1)} + \cdots + \dfrac{(-1)^{L+1}\gamma_{L+1}}{\hat{W}(\omega_{L+1})}} \tag{8.2.75}$$

where

$$\gamma_k = \prod_{\substack{n=0 \\ n\neq k}}^{L+1} \frac{1}{\cos\omega_k - \cos\omega_n} \tag{8.2.76}$$

The expression for δ in (8.2.75) follows immediately from the matrix equation in (8.2.74). Thus with an initial guess at the $L+2$ extremal frequencies, we compute δ.

Now since $P(\omega)$ is a trigometric polynomial of the form

$$P(\omega) = \sum_{k=0}^{L} \alpha(k) x^k \qquad x = \cos\omega$$

and since we know that the polynomial at the points $x_n \equiv \cos\omega_n$, $n = 0, 1, \ldots, L+1$, has the corresponding values

$$P(\omega_n) = \hat{H}_{dr}(\omega_n) - \frac{(-1)^n \delta}{\hat{W}(\omega_n)} \qquad n = 0, 1, \ldots, L+1 \tag{8.2.77}$$

we can use the Lagrange interpolation formula for $P(\omega)$. Thus $P(\omega)$ can be expressed as [see Hamming (1962)]

$$P(\omega) = \frac{\sum_{k=0}^{L} P(\omega_k)[\beta_k/(x - x_k)]}{\sum_{k=0}^{L} [\beta_k/(x - x_k)]} \tag{8.2.78}$$

where $P(\omega_n)$ is given by (8.2.77), $x = \cos\omega$, $x_k = \cos\omega_k$, and

$$\beta_k = \prod_{\substack{n=0 \\ n\neq k}}^{L} \frac{1}{x_k - x_n} \tag{8.2.79}$$

Having the solution for $P(\omega)$, we can now compute the error function $E(\omega)$ from

$$E(\omega) = \hat{W}(\omega)[\hat{H}_{dr}(\omega) - P(\omega)] \tag{8.2.80}$$

on a dense set of frequency points. Usually, a number of points equal to $16M$, where M is the length of the filter, suffices. If $|E(\omega)| \geq \delta$ for some frequencies on the dense set, then a new set of frequencies corresponding to the $L+2$ largest peaks of $|E(\omega)|$ are selected and the computational procedure beginning with (8.2.75) is repeated. Since the new set of $L + 2$ extremal frequencies are selected to correspond to the peaks of the error function $|E(\omega)|$, the algorithm forces δ to increase in each iteration until it converges to the upper bound and hence to the optimum solution for the Chebyshev approximation problem. In other words, when $|E(\omega)| \leq \delta$ for all frequencies on the dense set, the optimal solution has been found in terms of the polynomial $H(\omega)$.

A flowchart of the algorithm is shown in Fig. 8.18 and is due to Remez (1957).

Once the optimal solution has been obtained in terms of $P(\omega)$, the unit sample response $h(n)$ can be computed directly, without having to compute the parameters $\{\alpha(k)\}$. In effect, we have determined

$$H_r(\omega) = Q(\omega)P(\omega)$$

which can be evaluated at $\omega = 2\pi k/M$, $k = 0, 1, \ldots, (M - 1)/2$, for M odd, or $M/2$ for M even. Then, depending on the type of filter being designed, $h(n)$ can be determined from the formulas given in Table 8.3.

A computer program written by Parks and McClellan (1972b) is available for designing linear phase FIR filters based on the Chebyshev approximation criterion and implemented with the Remez exchange algorithm. This program can be used to design lowpass, highpass or bandpass filters, differentiators, and Hilbert transformers. The latter two types of filters are described in the following sections. A number of software packages for designing equiripple linear-phase FIR filters are now available.

The Parks–McClellan program requires a number of input parameters which determine the filter characteristics. In particular, the following parameters must

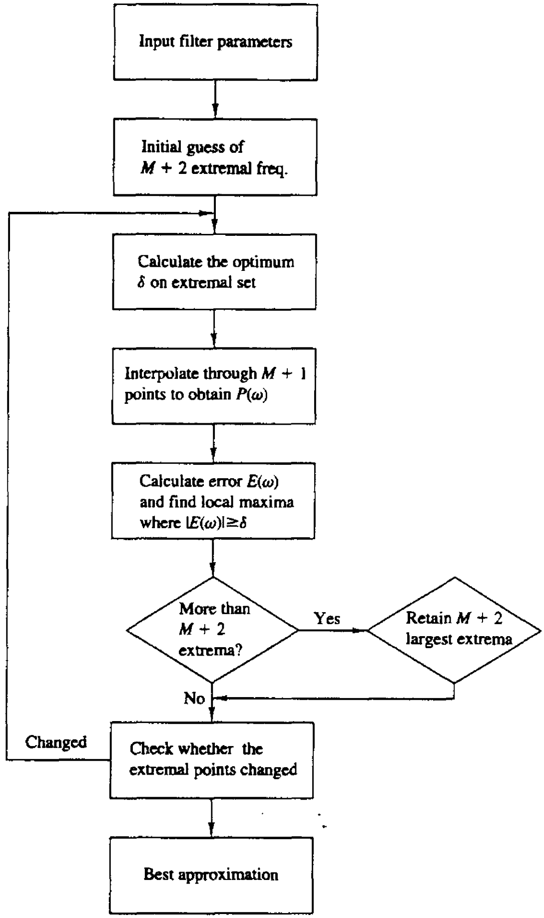

Figure 8.18 Flowchart of Remez algorithm.

be specified:

LINE 1

NFILT: The filter length, denoted above as M.

JTYPE: Type of filter:
JTYPE = 1 results in a multiple passband/stopband filter.

JTYPE = 2 results in a differentiator.
JTYPE = 3 results in a Hilbert transformer.

NBANDS: The number of frequency bands from 2 (for a lowpass filter) to a maximum of 10 (for a multiple-band filter).

LGRID: The grid density for interpolating the error function $E(\omega)$. The default value is 16 if left unspecified.

LINE 2

EDGE: The frequency bands specified by lower and upper cutoff frequencies, up to a maximum of 10 bands (an array of size 20, maximum). The frequencies are given in terms of the variable $f = \omega/2\pi$, where $f = 0.5$ corresponds to the folding frequency.

LINE 3

FX: An array of maximum size 10 that specifies the desired frequency response $H_{dr}(\omega)$ in each band.

LINE 4

WTX: An array of maximum size 10 that specifies the weight function in each band.

The following examples demonstrate the use of this program to design a lowpass and a bandpass filter.

Example 8.2.3

Design a lowpass filter of length $M = 61$ with a passband edge frequency $f_p = 0.1$ and a stopband edge frequency $f_s = 0.15$.

Solution The lowpass filter is a two-band filter with passband edge frequencies $(0, 0.1)$ and stopband edge frequencies $(0.15, 0.5)$. The desired response is $(1, 0)$ and the weight function is arbitrarily selected as $(1, 1)$.

61, 1, 2
0.0, 0.1, 0.15, 0.5
1.0, 0.0
1.0, 1.0

The result of this design is illustrated in Table 8.6, which gives the filter coefficients. The frequency response is shown in Fig. 8.19. The resulting filter has a stopband attenuation of −56 dB and a passband ripple of 0.0135 dB.

If we increase the length of the filter to $M = 101$ while maintaining all the other parameters given above the same, the resulting filter has the frequency response characteristic shown in Fig. 8.20. Now, the stopband attenuation is −85 dB and the passband ripple is reduced to 0.00046 dB.

We should indicate that it is possible to increase the attenuation in the stopband by keeping the filter length fixed, say at $M = 61$, and decreasing the weighting function $W(\omega) = \delta_2/\delta_1$ in the passband. With $M = 61$ and a weighting function

TABLE 8.6 PARAMETERS FOR LOWPASS FILTER DESIGN IN EXAMPLE 8.2.3

```
                 FINITE IMPULSE RESPONSE (FIR)
               LINEAR PHASE DIGITAL FILTER DESIGN
                   REMEZ EXCHANGE ALGORITHM
                      FILTER LENGTH = 61
                ***** IMPULSE RESPONSE *****
              H( 1) = -0.12109351E-02 = H( 61)
              H( 2) = -0.67270687E-03 = H( 60)
              H( 3) =  0.98090240E-04 = H( 59)
              H( 4) =  0.13536664E-02 = H( 58)
              H( 5) =  0.22969784E-02 = H( 57)
              H( 6) =  0.19963495E-02 = H( 56)
              H( 7) =  0.97026095E-04 = H( 55)
              H( 8) = -0.26466695E-02 = H( 54)
              H( 9) = -0.45133103E-02 = H( 53)
              H(10) = -0.37704944E-02 = H( 52)
              H(11) =  0.13079655E-04 = H( 51)
              H(12) =  0.51791356E-02 = H( 50)
              H(13) =  0.84883478E-02 = H( 49)
              H(14) =  0.69532110E-02 = H( 48)
              H(15) =  0.71037059E-04 = H( 47)
              H(16) = -0.90407897E-02 = H( 46)
              H(17) = -0.14723047E-01 = H( 45)
              H(18) = -0.11958945E-01 = H( 44)
              H(19) = -0.29799214E-04 = H( 43)
              H(20) =  0.15713422E-01 = H( 42)
              H(21) =  0.25657151E-01 = H( 41)
              H(22) =  0.21057373E-01 = H( 40)
              H(23) =  0.68637768E-04 = H( 39)
              H(24) = -0.28902054E-01 = H( 38)
              H(25) = -0.49118541E-01 = H( 37)
              H(26) = -0.42713970E-01 = H( 36)
              H(27) = -0.50114304E-04 = H( 35)
              H(28) =  0.73574215E-01 = H( 34)
              H(29) =  0.15782040E+00 = H( 33)
              H(30) =  0.22465512E+00 = H( 32)
              H(31) =  0.25007001E+00 = H( 31)
                            BAND  1        BAND  2
      LOWER BAND EDGE     0.0000000      0.1500000
      UPPER BAND EDGE     0.1000000      0.5000000
      DESIRED VALUE       1.0000000      0.0000000
      WEIGHTING           1.0000000      1.0000000
      DEVIATION           0.0015537      0.0015537
      DEVIATION IN DB     0.0134854    -56.1724014
EXTREMAL FREQUENCIES--MAXIMA OF THE ERROR CURVE
0.0000000   0.0252016   0.0423387   0.0584677   0.0735887
0.0866935   0.0957661   0.1000000   0.1500000   0.1540323
0.1631048   0.1762097   0.1903225   0.2054435   0.2215725
0.2377015   0.2538306   0.2699596   0.2860886   0.3022176
0.3183466   0.3354837   0.3516127   0.3677417   0.3848788
0.4010078   0.4171368   0.4342739   0.4504029   0.4665320
0.4836690   0.5000000
```

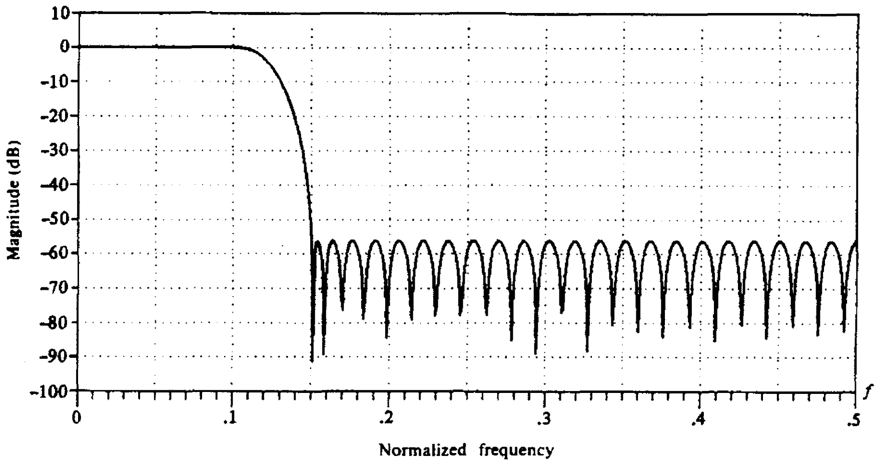

Figure 8.19 Frequency response of $M = 61$ FIR filter in Example 8.2.3.

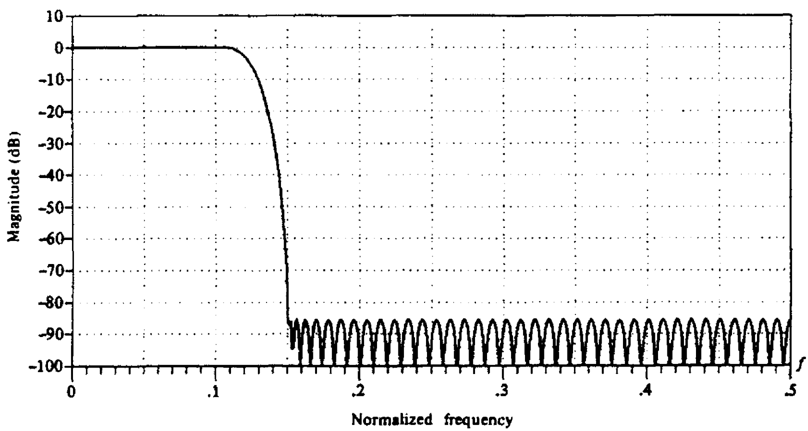

Figure 8.20 Frequency response of $M = 101$ FIR filter in Example 8.2.3.

$(0.1, 1)$, we obtain a filter that has a stopband attenuation of -65 dB and a passband ripple of 0.049 dB.

Example 8.2.4

Design a bandpass filter of length $M = 32$ with passband edge frequencies $f_{p1} = 0.2$ and $f_{p2} = 0.35$ and stopband edge frequencies of $f_{s1} = 0.1$ and $f_{s2} = 0.425$.

Solution This passband filter is a three-band filter with a stopband range of (0, 0.1), a passband range of (0.2, 0.35), and a second stopband range of (0.425, 0.5). The weighting function is selected as (10.0, 1.0, 10.0), or as (1.0, 0.1, 1.0), and the desired response in the three bands is (0.0, 1.0, 0.0). Thus the input parameters to the program are

32, 1, 3
0.0, 0.1, 0.2, 0.35, 0.425, 0.5
0.0, 1.0, 0.0
10.0, 1.0, 10.0

The results of this design are shown in Table 8.7, which gives the filter coefficients. We note that the ripple in the stopbands δ_2 is 10 times smaller than the ripple in

TABLE 8.7 PARAMETERS FOR BANDPASS FILTER IN EXAMPLE 8.2.4

```
                    FINITE IMPULSE RESPONSE (FIR)
                  LINEAR PHASE DIGITAL FILTER DESIGN
                      REMEZ EXCHANGE ALGORITHM
                          BANDPASS FILTER
                         FILTER LENGTH = 32
                   ***** IMPULSE RESPONSE *****
                  H( 1) = -0.57534026E-02 = H( 32)
                  H( 2) =  0.99026691E-03 = H( 31)
                  H( 3) =  0.75733471E-02 = H( 30)
                  H( 4) = -0.65141204E-02 = H( 29)
                  H( 5) =  0.13960509E-01 = H( 28)
                  H( 6) =  0.22951644E-02 = H( 27)
                  H( 7) = -0.19994041E-01 = H( 26)
                  H( 8) =  0.71369656E-02 = H( 25)
                  H( 9) = -0.39657373E-01 = H( 24)
                  H(10) =  0.11260066E-01 = H( 23)
                  H(11) =  0.66233635E-01 = H( 22)
                  H(12) = -0.10497202E-01 = H( 21)
                  H(13) =  0.85136160E-01 = H( 20)
                  H(14) = -0.12024988E+00 = H( 19)
                  H(15) = -0.29678580E+00 = H( 18)
                  H(16) =  0.30410913E+00 = H( 17)
                        BAND 1         BAND 2          BAND 3
LOWER BAND EDGE       0.0000000      0.2000000       0.4250000
UPPER BAND EDGE       0.1000000      0.3500000       0.5000000
DESIRED VALUE         0.0000000      1.0000000       0.0000000
WEIGHTING            10.0000000      1.0000000      10.0000000
DEVIATION             0.0015131      0.0151312       0.0015131
DEVIATION IN DB     -56.4025536      0.1304428     -56.4025536
EXTREMAL FREQUENCIES--MAXIMA OF THE ERROR CURVE
0.0000000    0.0273438    0.0527344    0.0761719    0.0937500
0.1000000    0.2000000    0.2195313    0.2527344    0.2839844
0.3132813    0.3386719    0.3500000    0.4250000    0.4328125
0.4503906    0.4796875
```

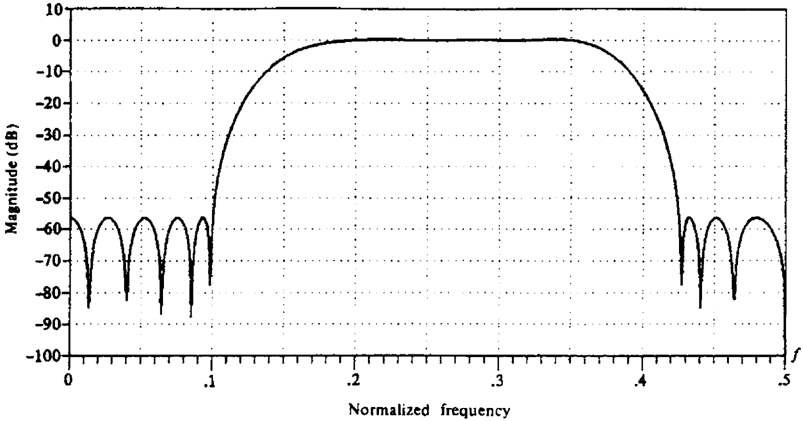

Figure 8.21 Frequency response of $M = 32$ FIR filter in Example 8.2.4.

the passband due to the fact that errors in the stopband were given a weight of 10 compared to the passband weight of unity. The frequency response of the bandpass filter is illustrated in Fig. 8.21.

These examples serve to illustrate the relative ease with which optimal lowpass, highpass, bandstop, bandpass, and more general multiband linear-phase FIR filters can be designed based on the Chebyshev approximation criterion implemented by means of the Remez exchange algorithm. In the next two sections we consider the design of differentiators and Hilbert transformers.

8.2.5 Design of FIR Differentiators

Differentiators are used in many analog and digital systems to take the derivative of a signal. An ideal differentiator has a frequency response that is linearly proportional to frequency. Similarly, an ideal digital differentiator is defined as one that has the frequency response

$$H_d(\omega) = j\omega \qquad -\pi \le \omega \le \pi \tag{8.2.81}$$

The unit sample response corresponding to $H_d(\omega)$ is

$$\begin{aligned} h_d(n) &= \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\omega)e^{j\omega n}\,d\omega \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} j\omega e^{j\omega n}\,d\omega \\ &= \frac{\cos \pi n}{n} \qquad -\infty < n < \infty, n \neq 0 \end{aligned} \tag{8.2.82}$$

We observe that the ideal differentiator has an antisymmetric unit sample response [i.e., $h_d(n) = -h_d(-n)$]. Hence, $h_d(0) = 0$.

In this section we consider the design of linear-phase FIR differentiators based on the Chebyshev approximation criterion. In view of the fact that the ideal differentiator has an antisymmetric unit sample response, we shall confine our attention to FIR designs in which $h(n) = -h(M-1-n)$. Hence we consider the filter types classified in the preceding section, as Case 3 and Case 4.

We recall that in Case 3, where M is odd, the real-valued frequency response of the FIR filter $H_r(\omega)$ has the characteristic that $H_r(0) = 0$. A zero response at zero frequency is just the condition that the differentiator should satisfy, and we see from Table 8.5 that both filter types satisfy this condition. However, if a full-band differentiator is desired, this is impossible to achieve with an FIR filter having an odd number of coefficients, since $H_r(\pi) = 0$ for M odd. In practice, however, full-band differentiators are rarely required.

In most cases of practical interest, the desired frequency response characteristic need only be linear over the limited frequency range $0 \le \omega \le 2\pi f_p$, where f_p is called the bandwidth of the differentiator. In the frequency range $2\pi f_p < \omega \le \pi$, the desired response may be either left unconstrained or constrained to be zero.

In the design of FIR differentiators based on the Chebyshev approximation criterion, the weighting function $W(\omega)$ is specified in the program as

$$W(\omega) = \frac{1}{\omega} \qquad 0 \le \omega \le 2\pi f_p \tag{8.2.83}$$

in order that the relative ripple in the passband be a constant. Thus the absolute error between the desired response ω and the approximation $H_r(\omega)$ increases as ω varies from 0 to $2\pi f_p$. However, the weighting function in (8.2.83) ensures that the relative error

$$\begin{aligned} \delta &= \max_{0\le\omega\le 2\pi f_p} \{W(\omega)[\omega - H_r(\omega)]\} \\ &= \max_{0\le\omega\le 2\pi f_p} \left[1 - \frac{H_r(\omega)}{\omega}\right] \end{aligned} \tag{8.2.84}$$

is fixed within the passband of the differentiator.

Example 8.2.5

Use the Remez algorithm to design a linear-phase FIR differentiator of length $M = 60$. The passband edge frequency is 0.1 and the stopband edge frequency is 0.15.

Solution The input parameters to the program are

60,	2,	2	
0.0,	0.1,	0.15,	0.5
1.0,	0.0		
1.0,	1.0		

The results of this design including the filter coefficients are shown in Table 8.8. The frequency response characteristic is illustrated in Fig. 8.22. Also shown in the same figure is the approximation error over the passband $0 \le f \le 0.1$ of the filter.

TABLE 8.8 PARAMETERS FOR FIR DIFFERENTIATOR IN EXAMPLE 8.2.5

```
            FINITE IMPULSE RESPONSE (FIR)
          LINEAR-PHASE DIGITAL FILTER DESIGN
              REMEZ EXCHANGE ALGORITHM
                  DIFFERENTIATOR
                 FILTER LENGTH = 60
           ***** IMPULSE RESPONSE *****
         H( 1) = -0.12478075E-02 = -H( 60)
         H( 2) = -0.15713560E-02 = -H( 59)
         H( 3) =  0.36846737E-02 = -H( 58)
         H( 4) =  0.19298020E-02 = -H( 57)
         H( 5) =  0.14264141E-02 = -H( 56)
         H( 6) = -0.17615277E-02 = -H( 55)
         H( 7) = -0.43110573E-02 = -H( 54)
         H( 8) = -0.46953405E-02 = -H( 53)
         H( 9) = -0.14105244E-02 = -H( 52)
         H(10) =  0.41694222E-02 = -H( 51)
         H(11) =  0.85736215E-02 = -H( 50)
         H(12) =  0.79813031E-02 = -H( 49)
         H(13) =  0.11833385E-02 = -H( 48)
         H(14) = -0.87396065E-02 = -H( 47)
         H(15) = -0.15401847E-01 = -H( 46)
         H(16) = -0.12878445E-01 = -H( 45)
         H(17) = -0.18826872E-03 = -H( 44)
         H(18) =  0.16620506E-01 = -H( 43)
         H(19) =  0.26741523E-01 = -H( 42)
         H(20) =  0.20892018E-01 = -H( 41)
         H(21) = -0.18584095E-02 = -H( 40)
         H(22) = -0.31109909E-01 = -H( 39)
         H(23) = -0.48822176E-01 = -H( 38)
         H(24) = -0.38673453E-01 = -H( 37)
         H(25) =  0.36760122E-02 = -H( 36)
         H(26) =  0.65462478E-01 = -H( 35)
         H(27) =  0.12066317E+00 = -H( 34)
         H(28) =  0.14182134E+00 = -H( 33)
         H(29) =  0.11403757E+00 = -H( 32)
         H(30) =  0.43620080E-01 = -H( 31)
                             BAND  1      BAND  2
    LOWER BAND EDGE        0.0000000    0.1500000
    UPPER BAND EDGE        0.1000000    0.5000000
    DESIRED SLOPE         10.0000000    0.0000000
    WEIGHTING              1.0000000    1.0000000
    DEVIATION              0.0073580    0.0073580
EXTREMAL FREQUENCIES---MAXIMA OF THE ERROR CURVE
0.0010417   0.0156250   0.0312500   0.0468750   0.0614583
0.0750000   0.0875000   0.0968750   0.1000000   0.1500000
0.1552083   0.1666667   0.1822916   0.1979166   0.2156249
0.2322916   0.2489582   0.2666668   0.2843754   0.3020839
0.3187508   0.3364594   0.3541680   0.3718765   0.3906268
0.4083354   0.4260439   0.4447942   0.4625027   0.4812530
0.5000000
```

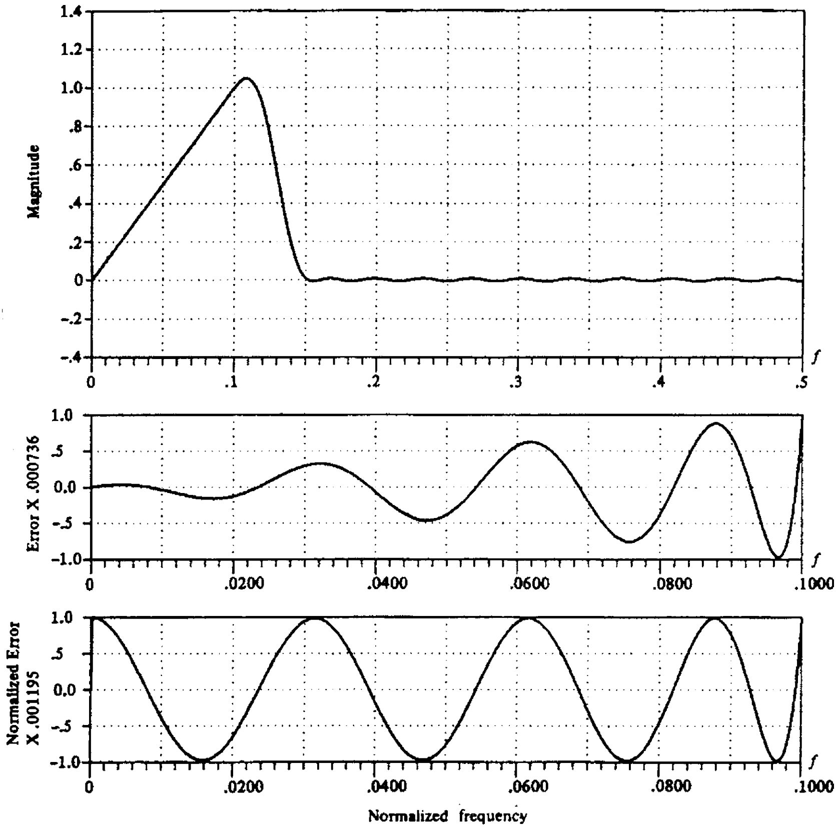

Figure 8.22 Frequency response and approximation error for $M = 60$ FIR differentiator of Example 8.2.5.

The important parameters in a differentiator are its length M, its bandwidth (band-edge frequency) f_p, and the peak relative error δ of the approximation. The interrelationship among these three parameters can be easily displayed parametrically. In particular, the value of $20\log_{10}\delta$ versus f_p with M as a parameter is shown in Fig. 8.23 for M even and in Fig. 8.24 for M odd. These results, due to Rabiner and Schafter (1974a), are useful in the selection of the filter length, given specifications on the inband ripple and the cutoff frequency f_p.

A comparison of the graphs in Figs. 8.23 and 8.24 reveals that even-length differentiators result in a significantly smaller approximation error δ than comparable odd-length differentiators. Designs based on M odd are particularly poor if

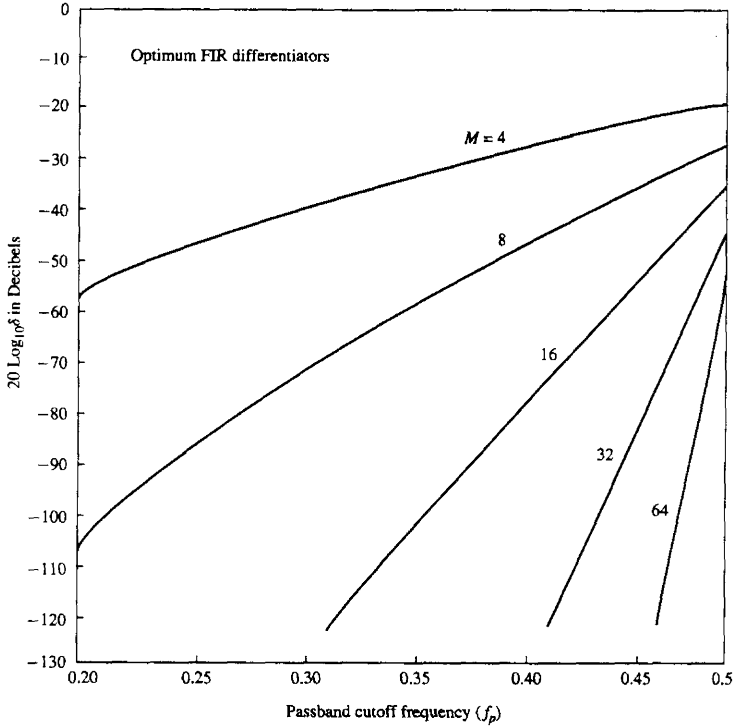

Figure 8.23 Curves of $20\log_{10}\delta$ versus f_p for $M = 4, 8, 16, 32$, and 64. [From paper by Rabiner and Schafer (1974a). Reprinted with permission of AT&T.]

the bandwidth exceeds $f_p = 0.45$. The problem is basically the zero in the frequency response at $\omega = \pi(f = 1/2)$. When $f_p < 0.45$, good designs are obtained for M odd, but comparable-length differentiators with M even are always better in the sense that the approximation error is smaller.

In view of the obvious advantage of even-length over odd-length differentiators, a conclusion might be that even-length differentiators are always preferable in practical systems. This is certainly true for many applications. However, we should note that the signal delay introduced by any linear-phase FIR filter is $(M-1)/2$, which is not an integer when M is even. In many practical applications, this is unimportant. In some applications where it is desirable to have an integer-valued delay in the signal at the output of the differentiator, we must select M to be odd.

These numerical results are based on designs resulting from the Chebyshev approximation criterion. We wish to indicate it is also possible and relatively

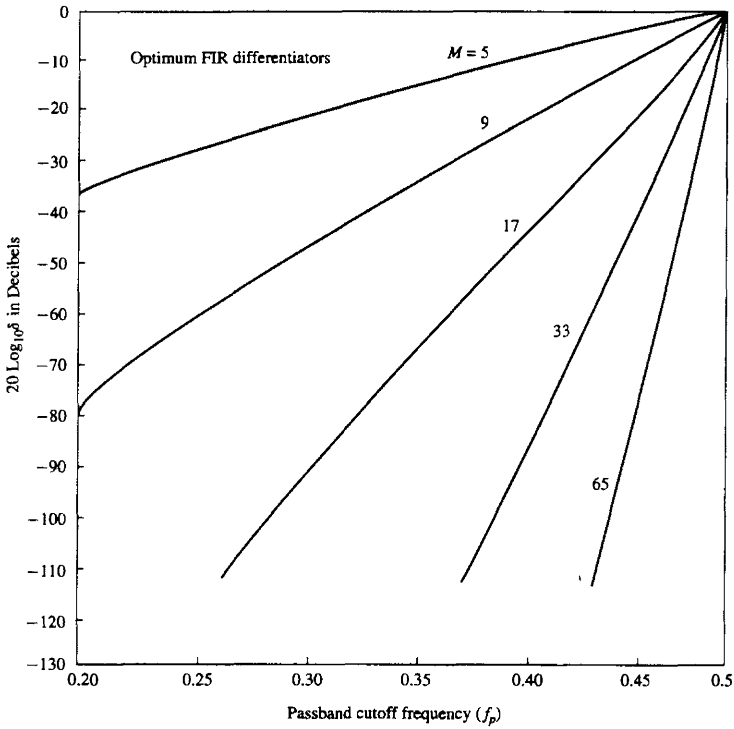

Figure 8.24 Curves of $20\log_{10}\delta$ versus F_p for $M = 5, 9, 17, 33$ and 65. [From paper by Rabiner and Schafer (1974a). Reprinted with permission of AT&T.]

easy to design linear phase FIR differentiators based on the frequency sampling method. For example, Fig. 8.25 illustrates the frequency response characteristics of a wideband ($f_p = 0.5$) differentiator, of length $M = 30$. The graph of the absolute value of the approximation error as a function of frequency is also shown in this figure.

8.2.6 Design of Hilbert Transformers

An ideal Hilbert transformer is an all-pass filter that imparts a 90° phase shift on the signal at its input. Hence the frequency response of the ideal Hilbert transformer is specified as

$$H_d(\omega) = \begin{cases} -j, & 0 < \omega \le \pi \\ j, & -\pi < \omega < 0 \end{cases} \tag{8.2.85}$$

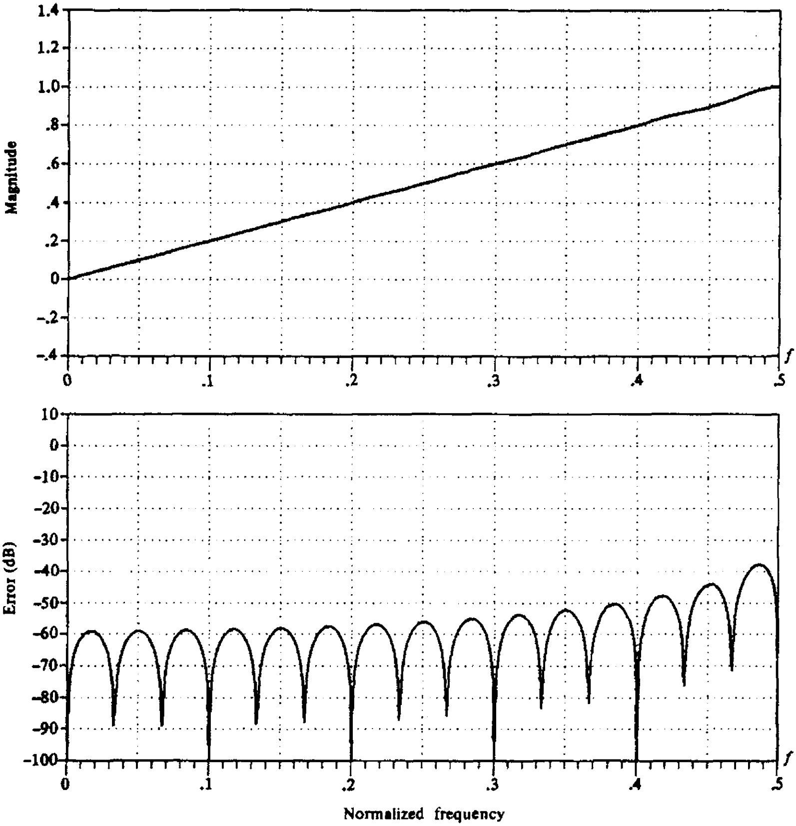

Figure 8.25 Frequency response and approximation error for $M = 30$ FIR differentiator designed by frequency sampling method.

Hilbert transformers are frequently used in communication systems and signal processing, as, for example, in the generation of single-sideband modulated signals, radar signal processing, and speech signal processing.

The unit sample response of an ideal Hilbert transformer is

$$
\begin{aligned}
h_d(n) &= \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\omega)e^{j\omega n}\,d\omega \\
&= \frac{1}{2\pi}\left(\int_{-\pi}^{0} je^{j\omega n}\,d\omega - \int_{0}^{\pi} je^{j\omega n}\,d\omega\right)
\end{aligned}
$$

$$= \begin{cases} \dfrac{2}{\pi}\dfrac{\sin^2(\pi n/2)}{n}, & n \neq 0 \\ 0, & n = 0 \end{cases} \tag{8.2.86}$$

As expected, $h_d(n)$ is infinite in duration and noncausal. We note that $h_d(n)$ is antisymmetric [i.e., $h_d(n) = -h_d(-n)$]. In view of this characteristic, we focus our attention on the design of linear-phase FIR Hilbert transformers with antisymmetric unit sample response [i.e., $h(n) = -h(M-1-n)$]. We also observe that our choice of an antisymmetric unit sample response is consistent with having a purely imaginary frequency response characteristic $H_d(\omega)$.

We recall once again that when $h(n)$ is antisymmetric, the real-valued frequency response characteristic $H_r(\omega)$ is zero at $\omega = 0$ for both M odd and even and at $\omega = \pi$ when M is odd. Clearly, then, it is impossible to design an all-pass digital Hilbert transformer. Fortunately, in practical signal processing applications, an all-pass Hilbert transformer is unnecessary. Its bandwidth need only cover the bandwidth of the signal to be phase shifted. Consequently, we specify the desired real-valued frequency response of a Hilbert transform filter as

$$H_{dr}(\omega) = 1 \qquad 2\pi f_l \leq \omega \leq 2\pi f_u \tag{8.2.87}$$

where f_l and f_u are the lower and upper cutoff frequencies, respectively.

It is interesting to note that the ideal Hilbert transformer with unit sample response $h_d(n)$ as given in (8.2.86) is zero for n even. This property is retained by the FIR Hilbert transformer under some symmetry conditions. In particular, let us consider the Case 3 filter type for which

$$H_r(\omega) = \sum_{k=1}^{(M-1)/2} c(k) \sin \omega k \tag{8.2.88}$$

and suppose that $f_l = 0.5 - f_u$. This ensures a symmetric passband about the midpoint frequency $f = 0.25$. If we have this symmetry in the frequency response, $H_r(\omega) = H_r(\pi - \omega)$ and hence (8.2.88) yields

$$\begin{aligned} \sum_{k=1}^{(M-1)/2} c(k) \sin \omega k &= \sum_{k=1}^{(M-1)/2} c(k) \sin k(\pi - \omega) \\ &= \sum_{k=1}^{(M-1)/2} c(k) \sin \omega k \cos \pi k \\ &= \sum_{k=1}^{(M-1)/2} c(k)(-1)^{k+1} \sin \omega k \end{aligned}$$

or equivalently,

$$\sum_{k=1}^{(M-1)/2} [1 - (-1)^{k+1}] c(k) \sin \omega k = 0 \tag{8.2.89}$$

Clearly, $c(k)$ must be equal to zero for $k = 0, 2, 4, \ldots$.

Now, the relationship between $\{c(k)\}$ and the unit sample response $\{h(n)\}$ is, from (8.2.54),

$$c(k) = 2h\left(\frac{M-1}{2} - k\right)$$

or, equivalently,

$$h\left(\frac{M-1}{2} - k\right) = \frac{1}{2}c(k) \tag{8.2.90}$$

If $c(k)$ is zero for $k = 0, 2, 4, \ldots$, then (8.2.90) yields

$$h(k) = \begin{cases} 0, & k = 0, 2, 4, \ldots, \text{ for } \dfrac{M-1}{2} \text{ even} \\ 0, & k = 1, 3, 5, \ldots, \text{ for } \dfrac{M-1}{2} \text{ odd} \end{cases} \tag{8.2.91}$$

Unfortunately, (8.2.91) holds only for M odd. It does not hold for M even. This means that for comparable values of M, the case M odd is preferable since the computational complexity (number of multiplications and additions per output point) is roughly one half of that for M even.

When the design of the Hilbert transformer is performed by the Chebyshev approximation criterion using the Remez algorithm, we select the filter coefficients to minimize the peak approximation error

$$\begin{aligned} \delta &= \max_{2\pi f_l \le \omega \le 2\pi f_u} [H_{dr}(\omega) - H_r(\omega)] \\ &= \max_{2\pi f_l \le \omega \le 2\pi f_u} [1 - H_r(\omega)] \end{aligned} \tag{8.2.92}$$

Thus the weighting function is set to unity and the optimization is performed over the single frequency band (i.e., the passband of the filter).

Example 8.2.6

Design a Hilbert transformer with parameters $M = 31$, $f_l = 0.05$, and $f_u = 0.45$.

Solution We observe that the frequency response is symmetric, since $f_u = 0.5 - f_l$. The parameters for executing the Remez algorithm are

```
  31,    3,  1
0.05,  0.45
 1.0
 1.0
```

The result of this design is the unit sample response coefficients and the peak approximation error $\delta = 0.0026803$ or -51.4 dB given in Table 8.9. We observe that, indeed, every other value of $h(n)$ is essentially zero (these values are of the order of 10^{-7}). The frequency response of the Hilbert transformer is shown in Fig. 8.26.

TABLE 8.9 PARAMETERS FOR FIR HILBERT TRANSFORM FILTER IN EXAMPLE 8.2.6

```
                 FINITE IMPULSE RESPONSE (FIR)
               LINEAR PHASE DIGITAL FILTER DESIGN
                   REMEZ EXCHANGE ALGORITHM
                     HILBERT TRANSFORMER
                      FILTER LENGTH = 31
                ***** IMPULSE RESPONSE *****
              H( 1) =  0.41957516E-02 = -H( 31)
              H( 2) =  0.64310257E-07 = -H( 30)
              H( 3) =  0.92822444E-02 = -H( 29)
              H( 4) =  0.52693927E-07 = -H( 28)
              H( 5) =  0.18835988E-01 = -H( 27)
              H( 6) =  0.82308283E-07 = -H( 26)
              H( 7) =  0.34401190E-01 = -H( 25)
              H( 8) =  0.93328794E-07 = -H( 24)
              H( 9) =  0.59551738E-01 = -H( 23)
              H(10) =  0.50821171E-07 = -H( 22)
              H(11) =  0.10303782E+00 = -H( 21)
              H(12) =  0.17612138E-07 = -H( 20)
              H(13) =  0.19683167E+00 = -H( 19)
              H(14) = -0.23977606E-07 = -H( 18)
              H(15) =  0.63135374E+00 = -H( 17)
              H(16) =  0.0
                                     BAND 1
                LOWER BAND EDGE     0.0500000
                UPPER BAND EDGE     0.4500000
                DESIRED VALUE       1.0000000
                WEIGHTING           1.0000000
                DEVIATION           0.0026803
EXTREMAL FREQUENCIES---MAXIMA OF THE ERROR CURVE
0.0500000   0.0562500   0.0750000   0.1000000   0.1291666
0.1583333   0.1874999   0.2187499   0.2499998   0.2812498
0.3124998   0.3416664   0.3708331   0.3999997   0.4249997
0.4437497
```

Rabiner and Schafer (1974b) have investigated the characteristics of Hilbert transformer designs for both M odd and M even. If the filter design is restricted to a symmetric frequency response, then there are basically three parameters of interest, M, δ, and f_l. Figure 8.27 is a plot of $20\log_{10}\delta$ versus f_l with M as a parameter. We observe that for comparable values of M, there is no performance advantage of using M odd over M even, and vice versa. However, the computational complexity in implementing a filter for M odd is less by a factor of 2 over M even as previously indicated. Therefore, M odd is preferable in practice.

For design purposes, the graphs in Fig. 8.27 suggest that, as a rule of thumb,

$$Mf_l \approx -0.61\log_{10}\delta \tag{8.2.93}$$

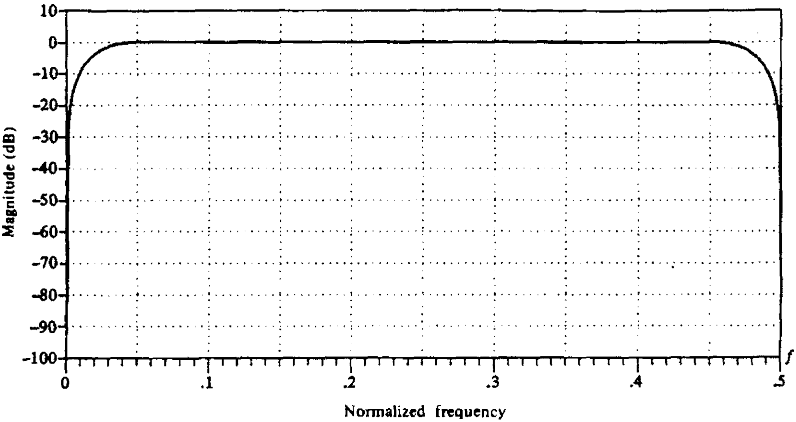

Figure 8.26 Frequency response of FIR Hilbert transform filter in Example 8.6.6.

Hence this formula can be used to estimate the size of one of the three basic filter parameters when the other two parameters are specified.

In concluding this section, we wish to show that Hilbert transformers can also be designed by the window method and the frequency sampling method. For example, Fig. 8.28 illustrates the frequency response of an $M = 31$ Hilbert transformer designed using the frequency sampling method. The corresponding values of the unit sample response are given in Table 8.10. A comparison of these filter parameters with those given in Table 8.9 indicates some small differences. In particular, it appears that the Chebyshev approximation criterion gives significantly smaller values for the filter coefficients that should be zero. In general, the Chebyshev approximation criterion results in better filter designs.

8.2.7 Comparison of Design Methods for Linear-Phase FIR Filters

Historically, the design method based on the use of windows to truncate the impulse response $h_d(n)$ and obtaining the desired spectral shaping, was the first method proposed for designing linear-phase FIR filters. The frequency-sampling method and the Chebyshev approximation method were developed in the 1970s and have since become very popular in the design of practical linear-phase FIR filters.

The major disadvantage of the window design method is the lack of precise control of the critical frequencies, such as ω_p and ω_s, in the design of a lowpass FIR filter. The values of ω_p and ω_s, in general, depend on the type of window and the filter length M.

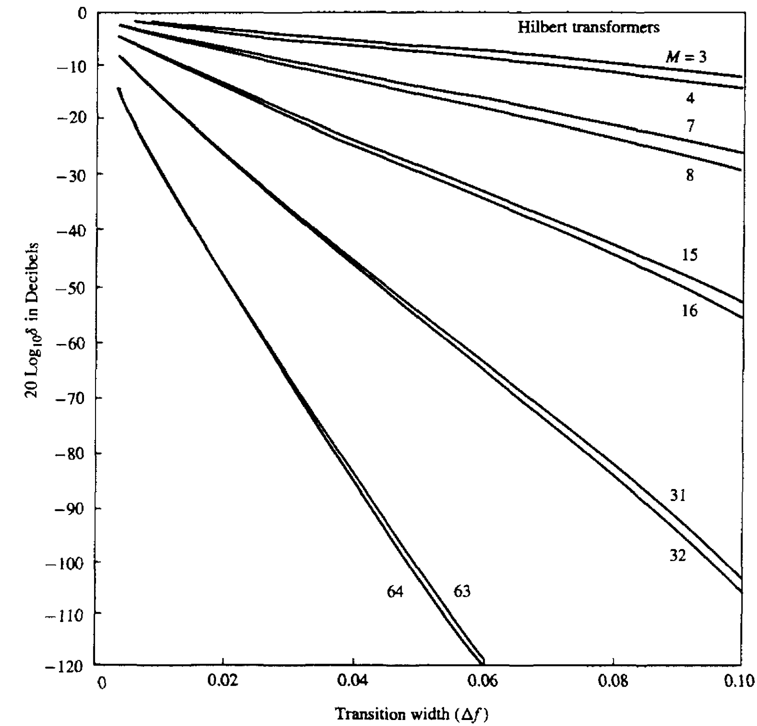

Figure 8.27 Curves of $20 \log_{10} \delta$ versus Δf for $M = 3, 4, 7, 8, 15, 16, 31, 32, 63, 64$. [From paper by Rabiner and Schafer (1974b). Reprinted with permission of AT&T.]

The frequency sampling method provides an improvement over the window design method, since $H_r(\omega)$ is specified at the frequencies $\omega_k = 2\pi k/M$ or $\omega_k = \pi(2k+1)/M$ and the transition band is a multiple of $2\pi/M$. This filter design method is particularly attractive when the FIR filter is realized either in the frequency domain by means of the DFT or in any of the frequency sampling realizations. The attractive feature of these realizations is that $H_r(\omega_k)$ is either zero or unity at all frequencies, except in the transition band.

The Chebyshev approximation method provides total control of the filter specifications, and, as a consequence, it is usually preferable over the other two methods. For a lowpass filter, the specifications are given in terms of the parameters ω_p, ω_s, δ_1, δ_2, and M. We can specify the parameters ω_p, ω_s, M and δ, and optimize the filters relative to δ_2. By spreading the approximation error over the

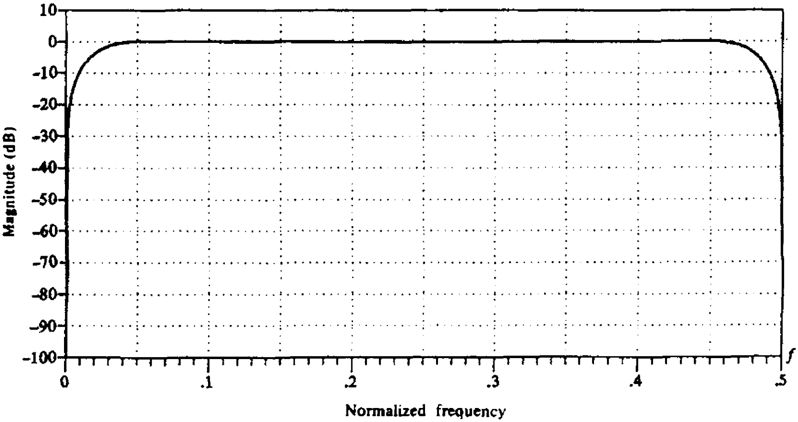

Figure 8.28 Frequency response of $M = 31$ FIR Hilbert transform filter designed by the frequency sampling method.

passband and the stopband of the filter, this method results in an optimal filter design, in the sense that for a given set of specifications just described, the maximum sidelobe level is minimized.

The Chebyshev design procedure based on the Remez exchange algorithm requires that we specify the length of the filter, the critical frequencies ω_p and ω_s, and the ratio δ_2/δ_1. However, it is more natural in filter design to specify ω_p, ω_s, δ_1, and δ_2 and to determine the filter length that satisfies the specifications. Although there is no simple formula to determine the filter length from these specifications, a number of approximations have been proposed for estimating M from ω_p, ω_s, δ_1, and δ_2. A particularly simple formula attributed to Kaiser for approximating M is

$$\hat{M} = \frac{-20\log_{10}\left(\sqrt{\delta_1\delta_2}\right) - 13}{14.6\ \Delta f} + 1 \tag{8.2.94}$$

where Δf is the transition band, defined as $\Delta f = (\omega_s - \omega_p)/2\pi$. This formula has been given in the paper by Rabiner et al. (1975). A more accurate formula proposed by Herrmann et al. (1973) is

$$\hat{M} = \frac{D_\infty(\delta_1, \delta_2) - f(\delta_1, \delta_2)(\Delta f)^2}{\Delta f} + 1 \tag{8.2.95}$$

where, by definition,

$$\begin{aligned} D_\infty(\delta_1, \delta_2) = {} & [0.005309(\log_{10}\delta_1)^2 + 0.07114(\log_{10}\delta_1) - 0.4761](\log_{10}\delta_2) \\ & - [0.00266(\log_{10}\delta_1)^2 + 0.5941\log_{10}\delta_1 + 0.4278] \end{aligned} \tag{8.2.96}$$

$$f(\delta_1, \delta_2) = 11.012 + 0.51244(\log_{10}\delta_1 - \log_{10}\delta_2) \tag{8.2.97}$$

TABLE 8.10 PARAMETERS A FOR $M = 31$ HILBERT TRANSFORM FILTER DESIGNED BY THE FREQUENCY-SAMPLING METHOD

```
LINEAR PHASE FIR HILBERT TRANSFORM
FREQUENCY SAMPLING METHOD

FILTER LENGTH =31
LOWER CUTOFF FREQUENCY (RELATIVE) = 0.5000000E-01
UPPER CUTOFF FREQUENCY (RELATIVE) = 0.4500000E+00

IMPULSE RESPONSE:
                         H( 0) = -0.1342662E-03
                         H( 1) =  0.2133148E-02
                         H( 2) =  0.4848863E-02
                         H( 3) =  0.2286159E-02
                         H( 4) =  0.1423532E-01
                         H( 5) =  0.1517075E-02
                         H( 6) =  0.3001805E-01
                         H( 7) =  0.5263533E-03
                         H( 8) =  0.5574721E-01
                         H( 9) = -0.2281570E-03
                         H(10) =  0.1001032E+00
                         H(11) = -0.5338326E-03
                         H(12) =  0.1949848E+00
                         H(13) = -0.3994641E-03
                         H(14) =  0.6307253E+00
                         H(15) = -0.9335956E-06
                         H(16) = -0.6307245E+00
                         H(17) =  0.3996222E-03
                         H(18) = -0.1949853E+00
                         H(19) =  0.5341307E-03
                         H(20) = -0.1001035E+00
                         H(21) =  0.2285338E-03
                         H(22) = -0.5574735E-01
                         H(23) = -0.5263340E-03
                         H(24) = -0.3001794E-01
                         H(25) = -0.1517240E-02
                         H(26) = -0.1423557E-01
                         H(27) = -0.2285915E-02
                         H(28) = -0.4848215E-02
                         H(29) = -0.2133800E-02
                         H(30) =  0.1344162E-03
```

These formulas are extremely useful in obtaining a good estimate of the filter length required to achieve the given specifications Δf, δ_1, and δ_2. The estimate is used to carry out the design and if the resulting δ exceeds the specified δ_2, the length can be increased until we obtain a sidelobe level that meets the specifications.

8.3 DESIGN OF IIR FILTERS FROM ANALOG FILTERS

Just as in the design of FIR filters, there are several methods that can be used to design digital filters having an infinite-duration unit sample response. The techniques described in this section are all based on converting an analog filter into a digital filter. Analog filter design is a mature and well developed field, so it is not surprising that we begin the design of a digital filter in the analog domain and then convert the design into the digital domain.

An analog filter can be described by its system function.

$$H_a(s) = \frac{B(s)}{A(s)} = \frac{\sum_{k=0}^{M} \beta_k s^k}{\sum_{k=0}^{N} \alpha_k s^k} \tag{8.3.1}$$

where $\{\alpha_k\}$ and $\{\beta_k\}$ are the filter coefficients, or by its impulse response, which is related to $H_a(s)$ by the Laplace transform

$$H_a(s) = \int_{-\infty}^{\infty} h(t)e^{-st}\,dt \tag{8.3.2}$$

Alternatively, the analog filter having the rational system function $H(s)$ given in (8.3.1), can be described by the linear constant-coefficient differential equation

$$\sum_{k=0}^{N} \alpha_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} \beta_k \frac{d^k x(t)}{dt^k} \tag{8.3.3}$$

where $x(t)$ denotes the input signal and $y(t)$ denotes the output of the filter.

Each of these three equivalent characterizations of an analog filter leads to alternative methods for converting the filter into the digital domain, as will be described in Sections 8.3.1 through 8.3.4. We recall that an analog linear time-invariant system with system function $H(s)$ is stable if all its poles lie in the left half of the s-plane. Consequently, if the conversion technique is to be effective, it should possess the following desirable properties:

1. The $j\Omega$ axis in the s-plane should map into the unit circle in the z-plane. Thus there will be a direct relationship between the two frequency variables in the two domains.

2. The left-half plane (LHP) of the s-plane should map into the inside of the unit circle in the z-plane. Thus a stable analog filter will be converted to a stable digital filter.

We mentioned in the preceding section that physically realizable and stable IIR filters cannot have linear phase. Recall that a linear-phase filter must have a system function that satisfies the condition

$$H(z) = \pm z^{-N} H(z^{-1}) \tag{8.3.4}$$

where z^{-N} represents a delay of N units of time. But if this were the case, the filter would have a mirror-image pole outside the unit circle for every pole inside the unit circle. Hence the filter would be unstable. Consequently, a causal and stable IIR filter cannot have linear phase.

If the restriction on physical realizability is removed, it is possible to obtain a linear-phase IIR filter, at least in principle. This approach involves performing a time reversal of the input signal $x(n)$, passing $x(-n)$ through a digital filter $H(z)$, time-reversing the output of $H(z)$, and finally, passing the result through $H(z)$ again. This signal processing is computationally cumbersome and appears to offer no advantages over linear-phase FIR filters. Consequently, when an application requires a linear-phase filter, it should be an FIR filter.

In the design of IIR filters, we shall specify the desired filter characteristics for the magnitude response only. This does not mean that we consider the phase response unimportant. Since the magnitude and phase characteristics are related, as indicated in Section 8.1, we specify the desired magnitude characteristics and accept the phase response that is obtained from the design methodology.

8.3.1 IIR Filter Design by Approximation of Derivatives

One of the simplest methods for converting an analog filter into a digital filter is to approximate the differential equation in (8.3.3) by an equivalent difference equation. This approach is often used to solve a linear constant-coefficient differential equation numerically on a digital computer.

For the derivative $dy(t)/dt$ at time $t = nT$, we substitute the *backward difference* $[y(nT) - y(nT-1)]/T$. Thus

$$\begin{aligned}\left.\frac{dy(t)}{dt}\right|_{t=nT} &= \frac{y(nT) - y(nT-T)}{T} \\ &= \frac{y(n) - y(n-1)}{T}\end{aligned} \tag{8.3.5}$$

where T represents the sampling interval and $y(n) \equiv y(nT)$. The analog differentiator with output $dy(t)/dt$ has the system function $H(s) = s$, while the digital system that produces the output $[y(n) - y(n-1)]/T$ has the system function $H(z) = (1 - z^{-1})/T$. Consequently, as shown in Fig. 8.29, the frequency-domain

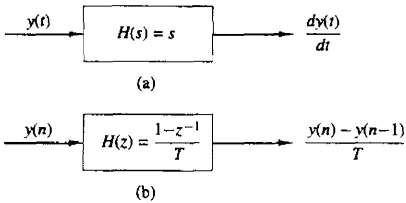

Figure 8.29 Substitution of the backward difference for the derivative implies the mapping $s = (1 - z^{-1})/T$.

equivalent for the relationship in (8.3.5) is

$$s = \frac{1 - z^{-1}}{T} \tag{8.3.6}$$

The second derivative $d^2y(t)/dt^2$ is replaced by the second difference, which is derived as follows:

$$\begin{aligned} \left.\frac{d^2y(t)}{dt^2}\right|_{t=nT} &= \frac{d}{dt}\left[\frac{dy(t)}{dt}\right]_{t=nT} \\ &= \frac{[y(nT) - y(nT - T)]/T - [y(nT - T) - y(nT - 2T)]/T}{T} \\ &= \frac{y(n) - 2y(n-1) + y(n-2)}{T^2} \end{aligned} \tag{8.3.7}$$

In the frequency domain, (8.3.7) is equivalent to

$$s^2 = \frac{1 - 2z^{-1} + z^{-2}}{T^2} = \left(\frac{1 - z^{-1}}{T}\right)^2 \tag{8.3.8}$$

It easily follows from the discussion that the substitution for the kth derivative of $y(t)$ results in the equivalent frequency-domain relationship

$$s^k = \left(\frac{1 - z^{-1}}{T}\right)^k \tag{8.3.9}$$

Consequently, the system function for the digital IIR filter obtained as a result of the approximation of the derivatives by finite differences is

$$H(z) = H_a(s)|_{s=(1-z^{-1})/T} \tag{8.3.10}$$

where $H_a(s)$ is the system function of the analog filter characterized by the differential equation given in (8.3.3).

Let us investigate the implications of the mapping from the s-plane to the z-plane as given by (8.3.6) or, equivalently,

$$z = \frac{1}{1 - sT} \tag{8.3.11}$$

If we substitute $s = j\Omega$ in (8.2.11), we find that

$$z = \frac{1}{1 - j\Omega T}$$

$$= \frac{1}{1+\Omega^2T^2} + j\frac{\Omega T}{1+\Omega^2T^2} \tag{8.3.12}$$

As Ω varies from $-\infty$ to ∞, the corresponding locus of points in the z-plane is a circle of radius $\frac{1}{2}$ and with center at $z = \frac{1}{2}$, as illustrated in Fig. 8.30.

It is easily demonstrated that the mapping in (8.3.11) takes points in the LHP of the s-plane into corresponding points inside this circle in the z-plane and points in the RHP of the s-plane are mapped into points outside this circle. Consequently, this mapping has the desirable property that a stable analog filter is transformed into a stable digital filter. However, the possible location of the poles of the digital filter are confined to relatively small frequencies and, as a consequence, the mapping is restricted to the design of lowpass filters and bandpass filters having relatively small resonant frequencies. It is not possible, for example, to transform a highpass analog filter into a corresponding highpass digital filter.

In an attempt to overcome the limitations in the mapping given above, more complex substitutions for the derivatives have been proposed. In particular, an Lth-order difference of the form

$$\left.\frac{dy(t)}{dt}\right|_{t=nT} = \frac{1}{T}\sum_{k=1}^{L}\alpha_k\frac{y(nT+kT)-y(nT-kT)}{T} \tag{8.3.13}$$

has been proposed, where $\{\alpha_k\}$ are a set of parameters that can be selected to optimize the approximation. The resulting mapping between the s-plane and the z-plane is now

$$s = \frac{1}{T}\sum_{k=1}^{L}\alpha_k(z^k - z^{-k}) \tag{8.3.14}$$

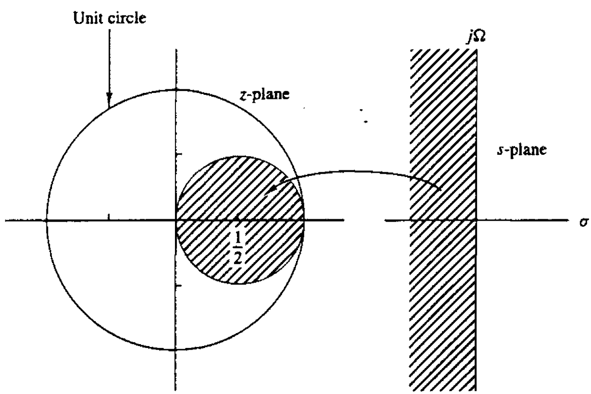

Figure 8.30 The mapping $s = (1 - z^{-1})/T$ takes LHP in the s-plane into points inside the circle of radius $\frac{1}{2}$ and center $z = \frac{1}{2}$ in the z-plane.

When $z = e^{j\omega}$, we have

$$s = j\frac{2}{T}\sum_{k=1}^{L} \alpha_k \sin \omega k \tag{8.3.15}$$

which is purely imaginary. Thus

$$\Omega = \frac{2}{T}\sum_{k=1}^{L} \alpha_k \sin \omega k \tag{8.3.16}$$

is the resulting mapping between the two frequency variables. By proper choice of the coefficients $\{\alpha_k\}$ it is possible to map the $j\Omega$-axis into the unit circle. Furthermore, points in the LHP in s can be mapped into points inside the unit circle in z.

Despite achieving the two desirable characteristics with the mapping of (8.3.16), the problem of selecting the set of coefficients $\{\alpha_k\}$ remains. In general, this is a difficult problem. Since simpler techniques exist for converting analog filters into IIR digital filters, we shall not emphasize the use of the Lth-order difference as a substitute for the derivative.

Example 8.3.1

Convert the analog bandpass filter with system function

$$H_a(s) = \frac{1}{(s+0.1)^2 + 9}$$

into a digital IIR filter by use of the backward difference for the derivative.

Solution Substitution for s from (8.3.6) into $H(s)$ yields

$$\begin{aligned} H(z) &= \frac{1}{\left(\dfrac{1-z^{-1}}{T} + 0.1\right)^2 + 9} \\ &= \frac{T^2/(1+0.2T+9.01T^2)}{1 - \dfrac{2(1+0.1T)}{1+0.2T+9.01T^2}z^{-1} + \dfrac{1}{1+0.2T+9.01T^2}z^{-2}} \end{aligned}$$

The system function $H(z)$ has the form of a resonator provided that T is selected small enough (e.g., $T \leq 0.1$), in order for the poles to be near the unit circle. Note that the condition $a_1^2 < 4a_2$ is satisfied, so that the poles are complex valued.

For example, if $T = 0.1$, the poles are located at

$$\begin{aligned} p_{1,2} &= 0.91 \pm j0.27 \\ &= 0.949e^{\pm j16.5^\circ} \end{aligned}$$

We note that the range of resonant frequencies is limited to low frequencies, due to the characteristics of the mapping. The reader is encouraged to plot the frequency response $H(\omega)$ of the digital filter for different values of T and compare the results with the frequency response of the analog filter.

Example 8.3.2

Convert the analog bandpass filter in Example 8.3.1 into a digital IIR filter by use of the mapping

$$s = \frac{1}{T}(z - z^{-1})$$

Solution By substituting for s in $H(s)$, we obtain

$$\begin{aligned} H(z) &= \frac{1}{\left(\dfrac{z - z^{-1}}{T} + 0.1\right)^2 + 9} \\ &= \frac{z^2T^2}{z^4 + 0.2Tz^3 + (2 + 9.01T^2)z^2 - 0.2Tz + 1} \end{aligned}$$

We observe that this mapping has introduced two additional poles in the conversion from $H_a(s)$ to $H(z)$. As a consequence, the digital filter is significantly more complex than the analog filter. This is a major drawback to the mapping given above.

8.3.2 IIR Filter Design by Impulse Invariance

In the impulse invariance method, our objective is to design an IIR filter having a unit sample response $h(n)$ that is the sampled version of the impulse response of the analog filter. That is,

$$h(n) \equiv h(nT) \qquad n = 0, 1, 2, \ldots \tag{8.3.17}$$

where T is the sampling interval.

To examine the implications of (8.3.17), we refer back to Section 4.2.9. Recall that when a continuous time signal $x_a(t)$ with spectrum $X_a(F)$ is sampled at a rate $F_s = 1/T$ samples per second, the spectrum of the sampled signal is the periodic repetition of the scaled spectrum $F_sX_a(F)$ with period F_s. Specifically, the relationship is

$$X(f) = F_s \sum_{k=-\infty}^{\infty} X_a[(f - k)F_s] \tag{8.3.18}$$

where $f = F/F_s$ is the normalized frequency. Aliasing occurs if the sampling rate F_s is less than twice the highest frequency contained in $X_a(F)$.

Expressed in the context of sampling the impulse response of an analog filter with frequency response $H_a(F)$, the digital filter with unit sample response $h(n) \equiv h_a(nT)$ has the frequency response

$$H(f) = F_s \sum_{k=-\infty}^{\infty} H_a[(f - k)F_s] \tag{8.3.19}$$

or, equivalently,

$$H(\omega) = F_s \sum_{k=-\infty}^{\infty} H_a[(\omega - 2\pi k)F_s] \tag{8.3.20}$$

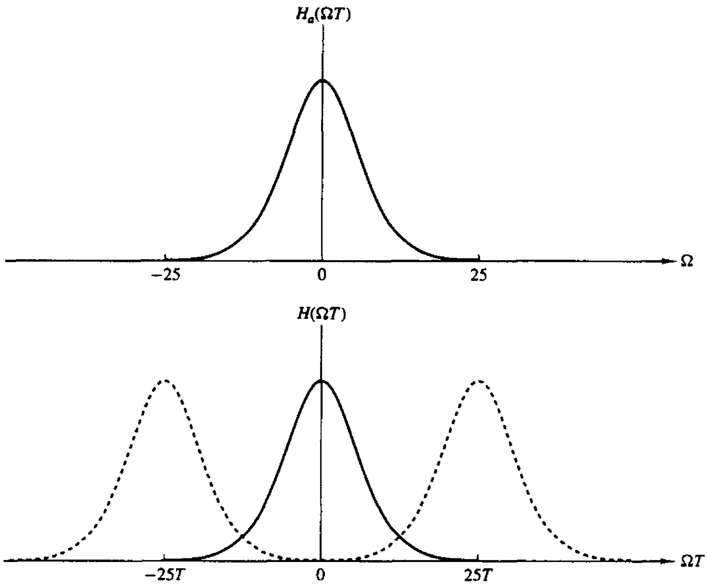

Figure 8.31 Frequency response $H_a(\Omega)$ of the analog filter and frequency response of the corresponding digital filter with aliasing.

or

$$H(\Omega T) = \frac{1}{T} \sum_{k=-\infty}^{\infty} H_a\left(\Omega - \frac{2\pi k}{T}\right) \tag{8.3.21}$$

Figure 8.31 depicts the frequency response of a lowpass analog filter and the frequency response of the corresponding digital filter.

It is clear that the digital filter with frequency response $H(\omega)$ has the frequency response characteristics of the corresponding analog filter if the sampling interval T is selected sufficiently small to completely avoid or at least minimize the effects of aliasing. It is also clear that the impulse invariance method is inappropriate for designing highpass filters due the to spectrum aliasing that results from the sampling process.

To investigate the mapping of points between the z-plane and the s-plane implied by the sampling process, we rely on a generalization of (8.3.21) which relates the z-transform of $h(n)$ to the Laplace transform of $h_a(t)$. This relationship is

$$H(z)|_{z=e^{sT}} = \frac{1}{T} \sum_{k=-\infty}^{\infty} H_a\left(s - j\frac{2\pi k}{T}\right) \tag{8.3.22}$$

where

$$H(z) = \sum_{n=0}^{\infty} h(n)z^{-n}$$

$$H(z)|_{z=e^{sT}} = \sum_{n=0}^{\infty} h(n)e^{-sTn} \tag{8.3.23}$$

Note that when $s = j\Omega$, (8.3.22) reduces to (8.3.21), where the factor of j in $H_a(\Omega)$ is suppressed in our notation.

Let us consider the mapping of points from the s-plane to the z-plane implied by the relation

$$z = e^{sT} \tag{8.3.24}$$

If we substitute $s = \sigma + j\Omega$ and express the complex variable z in polar form as $z = re^{j\omega}$, (8.3.24) becomes

$$re^{j\omega} = e^{\sigma T}e^{j\Omega T}$$

Clearly, we must have

$$\begin{aligned} r &= e^{\sigma T} \\ \omega &= \Omega T \end{aligned} \tag{8.3.25}$$

Consequently, $\sigma < 0$ implies that $0 < r < 1$ and $\sigma > 0$ implies that $r > 1$. When $\sigma = 0$, we have $r = 1$. Therefore, the LHP in s is mapped inside the unit circle in z and the RHP in s is mapped outside the unit circle in z.

Also, the $j\Omega$-axis is mapped into the unit circle in z as indicated above. However, the mapping of the $j\Omega$-axis into the unit circle is not one-to-one. Since ω is unique over the range $(-\pi, \pi)$, the mapping $\omega = \Omega T$ implies that the interval $-\pi/T \le \Omega \le \pi/T$ maps into the corresponding values of $-\pi \le \omega \le \pi$. Furthermore, the frequency interval $\pi/T \le \Omega \le 3\pi/T$ also maps into the interval $-\pi \le \omega \le \pi$ and, in general, so does the interval $(2k-1)\pi/T \le \Omega \le (2k+1)\pi/T$, when k is an integer. Thus the mapping from the analog frequency Ω to the frequency variable ω in the digital domain is many-to-one, which simply reflects the effects of aliasing due to sampling. Figure 8.32 illustrates the mapping from the s-plane to the z-plane for the relation in (8.3.24).

To explore further the effect of the impulse invariance design method on the characteristics of the resulting filter, let us express the system function of the analog filter in partial-fraction form. On the assumption that the poles of the analog filter are distinct, we can write

$$H_a(s) = \sum_{k=1}^{N} \frac{c_k}{s - p_k} \tag{8.3.26}$$

where $\{p_k\}$ are the poles of the analog filter and $\{c_k\}$ are the coefficients in the partial-fraction expansion. Consequently,

$$h_a(t) = \sum_{k=1}^{N} c_k e^{p_k t} \qquad t \ge 0 \tag{8.3.27}$$

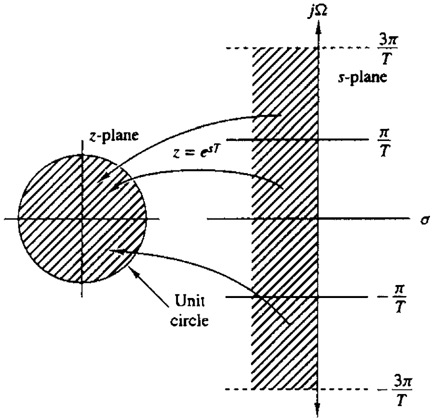

Figure 8.32 The mapping of $z = e^{sT}$ maps strips of width $2\pi/T$ (for $\sigma < 0$) in the s-plane into points in the unit circle in the z-plane.

If we sample $h_a(t)$ periodically at $t = nT$, we have

$$\begin{aligned} h(n) &= h_a(nT) \\ &= \sum_{k=1}^{N} c_k e^{p_k T n} \end{aligned} \tag{8.3.28}$$

Now, with the substitution of (8.3.28), the system function of the resulting digital IIR filter becomes

$$\begin{aligned} H(z) &= \sum_{n=0}^{\infty} h(n) z^{-n} \\ &= \sum_{n=0}^{\infty} \left(\sum_{k=1}^{N} c_k e^{p_k T n} \right) z^{-n} \\ &= \sum_{k=1}^{N} c_k \sum_{n=0}^{\infty} (e^{p_k T} z^{-1})^n \end{aligned} \tag{8.3.29}$$

The inner sum in (8.3.29) converges because $p_k < 0$ and yields

$$\sum_{n=0}^{\infty} (e^{p_k T} z^{-1})^n = \frac{1}{1 - e^{p_k T} z^{-1}} \tag{8.3.30}$$

Therefore, the system function of the digital filter is

$$H(z) = \sum_{k=1}^{N} \frac{c_k}{1 - e^{p_k T} z^{-1}} \tag{8.3.31}$$

We observe that the digital filter has poles at

$$z_k = e^{p_k T} \qquad k = 1, 2, \ldots, N \tag{8.3.32}$$

Although the poles are mapped from the s-plane to the z-plane by the relationship in (8.3.32), we should emphasize that the zeros in the two domains do not satisfy the same relationship. Therefore, the impulse invariance method does not correspond to the simple mapping of points given by (8.3.24).

The development that resulted in $H(z)$ given by (8.3.31) was based on a filter having distinct poles. It can be generalized to include multiple-order poles. For brevity, however, we shall not attempt to generalize (8.3.31).

Example 8.3.3

Convert the analog filter with system function

$$H_a(s) = \frac{s+0.1}{(s+0.1)^2+9}$$

into a digital IIR filter by means of the impulse invariance method.

Solution We note that the analog filter has a zero at $s = -0.1$ and a pair of complex-conjugate poles at

$$p_k = -0.1 \pm j3$$

as illustrated in Fig. 8.33.

We do not have to determine the impulse response $h_a(t)$ in order to design the digital IIR filter based on the method of impulse invariance. Instead, we directly determine $H(z)$, as given by (8.2.31), from the partial-fraction expansion of $H_a(s)$. Thus we have

$$H(s) = \frac{\frac{1}{2}}{s+0.1-j3} + \frac{\frac{1}{2}}{s+0.1+j3}$$

Then

$$H(z) = \frac{\frac{1}{2}}{1-e^{-0.1T}e^{j3T}z^{-1}} + \frac{\frac{1}{2}}{1-e^{-0.1T}e^{-j3T}z^{-1}}$$

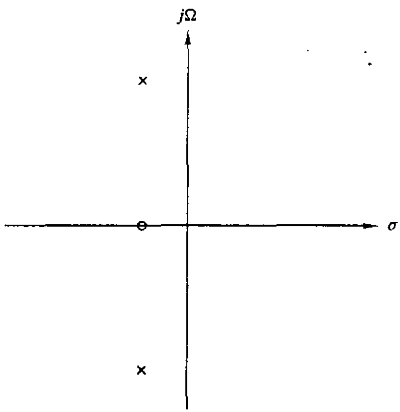

Figure 8.33 Pole–zero locations for analog filter in Example 8.3.3.

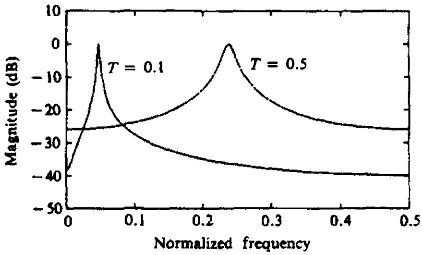

Figure 8.34 Frequency response of digital filter in Example 8.3.3.

Since the two poles are complex conjugates, we can combine them to form a single two-pole filter with system function

$$H(z) = \frac{1 - (e^{-0.1T} \cos 3T)z^{-1}}{1 - (2e^{-0.1T} \cos 3T)z^{-1} + e^{-0.2T}z^{-1}}$$

The magnitude of the frequency response characteristic of this filter is plotted in Fig. 8.34 for $T = 0.1$ and $T = 0.5$. For purpose of comparison, we have also plotted the magnitude of the frequency response of the analog filter in Fig. 8.35. We note that aliasing is significantly more prevalent when $T = 0.5$ than when $T = 0.1$. Also, note the shift of the resonant frequency as T changes.

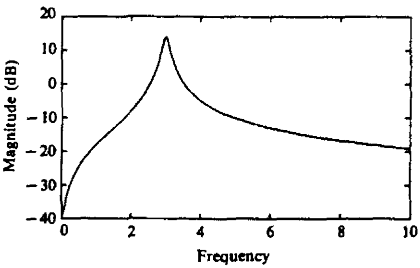

Figure 8.35 Frequency response of analog filter in Example 8.3.3.

The preceding example illustrates the importance of selecting a small value for T to minimize the effect of aliasing. Due to the presence of aliasing, the impulse invariance method is appropriate for the design of lowpass and bandpass filters only.

8.3.3 IIR Filter Design by the Bilinear Transformation

The IIR filter design techniques described in the preceding two sections have a severe limitation in that they are appropriate only for lowpass filters and a limited class of bandpass filters.

In this section we describe a mapping from the s-plane to the z-plane, called the bilinear transformation, that overcomes the limitation of the other two design methods described previously. The bilinear transformation is a conformal mapping that transforms the $j\Omega$-axis into the unit circle in the z-plane only once, thus avoiding aliasing of frequency components. Furthermore, all points in the LHP of s are mapped inside the unit circle in the z-plane and all points in the RHP of s are mapped into corresponding points outside the unit circle in the z-plane.

The bilinear transformation can be linked to the trapezoidal formula for numerical integration. For example, let us consider an analog linear filter with system function

$$H(s) = \frac{b}{s+a} \tag{8.3.33}$$

This system is also characterized by the differential equation

$$\frac{dy(t)}{dt} + ay(t) = bx(t) \tag{8.3.34}$$

Instead of substituting a finite difference for the derivative, suppose that we integrate the derivative and approximate the integral by the trapezoidal formula. Thus

$$y(t) = \int_{t_0}^{t} y'(\tau)d\tau + y(t_0) \tag{8.3.35}$$

where $y'(t)$ denotes the derivative of $y(t)$. The approximation of the integral in (8.3.35) by the trapezoidal formula at $t = nT$ and $t_0 = nT - T$ yields

$$y(nT) = \frac{T}{2}[y'(nT) + y'(nT - T)] + y(nT - T) \tag{8.3.36}$$

Now the differential equation in (8.3.34) evaluated at $t = nT$ yields

$$y'(nT) = -ay(nT) + bx(nT) \tag{8.3.37}$$

We use (8.3.37) to substitute for the derivative in (8.3.36) and thus obtain a difference equation for the equivalent discrete-time system. With $y(n) \equiv y(nT)$ and $x(n) \equiv x(nT)$, we obtain the result

$$\left(1 + \frac{aT}{2}\right) y(n) - \left(1 - \frac{aT}{2}\right) y(n-1) = \frac{bT}{2}[x(n) + x(n-1)] \tag{8.3.38}$$

The z-transform of this difference equation is

$$\left(1 + \frac{aT}{2}\right) Y(z) - \left(1 - \frac{aT}{2}\right) z^{-1}Y(z) = \frac{bT}{2}(1 + z^{-1})X(z)$$

Consequently, the system function of the equivalent digital filter is

$$H(z) = \frac{Y(z)}{X(z)} = \frac{(bT/2)(1 + z^{-1})}{1 + aT/2 - (1 - aT/2)z^{-1}}$$

or, equivalently,

$$H(z) = \frac{b}{\dfrac{2}{T}\left(\dfrac{1 - z^{-1}}{1 + z^{-1}}\right) + a} \tag{8.3.39}$$

Clearly, the mapping from the s-plane to the z-plane is

$$s = \frac{2}{T}\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right) \tag{8.3.40}$$

This is called the *bilinear transformation*.

Although our derivation of the bilinear transformation was performed for a first-order differential equation, it holds, in general, for an Nth-order differential equation.

To investigate the characteristics of the bilinear transformation, let

$$z = re^{j\omega}$$

$$s = \sigma + j\Omega$$

Then (8.3.40) can be expressed as

$$\begin{aligned} s &= \frac{2}{T}\frac{z-1}{z+1} \\ &= \frac{2}{T}\frac{re^{j\omega}-1}{re^{j\omega}+1} \\ &= \frac{2}{T}\left(\frac{r^2-1}{1+r^2+2r\cos\omega} + j\frac{2r\sin\omega}{1+r^2+2r\cos\omega}\right) \end{aligned}$$

Consequently,

$$\sigma = \frac{2}{T}\frac{r^2-1}{1+r^2+2r\cos\omega} \tag{8.3.41}$$

$$\Omega = \frac{2}{T}\frac{2r\sin\omega}{1+r^2+2r\cos\omega} \tag{8.3.42}$$

First, we note that if $r < 1$, then $\sigma < 0$, and if $r > 1$, then $\sigma > 0$. Consequently, the LHP in s maps into the inside of the unit circle in the z-plane and the RHP in s maps into the outside of the unit circle. When $r = 1$, then $\sigma = 0$ and

$$\begin{aligned} \Omega &= \frac{2}{T}\frac{\sin\omega}{1+\cos\omega} \\ &= \frac{2}{T}\tan\frac{\omega}{2} \end{aligned} \tag{8.3.43}$$

or, equivalently,

$$\omega = 2\tan^{-1}\frac{\Omega T}{2} \tag{8.3.44}$$

The relationship in (8.3.44) between the frequency variables in the two domains is illustrated in Fig. 8.36. We observe that the entire range in Ω is mapped only once into the range $-\pi \le \omega \le \pi$. However, the mapping is highly nonlinear. We observe a frequency compression or *frequency warping*, as it is usually called, due to the nonlinearity of the arctangent function.

It is also interesting to note that the bilinear transformation maps the point $s = \infty$ into the point $z = -1$. Consequently, the single-pole lowpass filter in

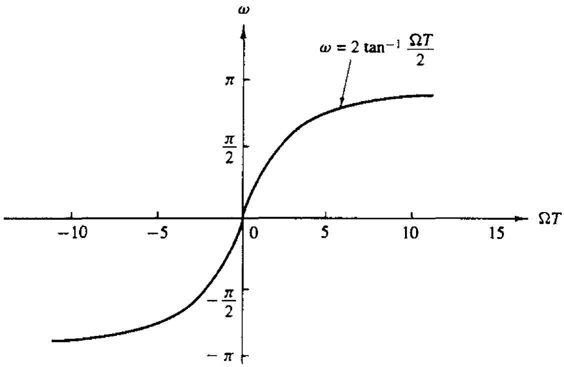

Figure 8.36 Mapping between the frequency variables ω and Ω resulting from the bilinear transformation.

(8.3.33), which has a zero at $s = \infty$, results in a digital filter that has a zero at $z = -1$.

Example 8.3.4

Convert the analog filter with system function

$$H_a(s) = \frac{s + 0.1}{(s + 0.1)^2 + 16}$$

into a digital IIR filter by means of the bilinear transformation. The digital filter is to have a resonant frequency of $\omega_r = \pi/2$.

Solution First, we note that the analog filter has a resonant frequency $\Omega_r = 4$. This frequency is to be mapped into $\omega_r = \pi/2$ by selecting the value of the parameter T. From the relationship in (8.3.43), we must select $T = \frac{1}{2}$ in order to have $\omega_r = \pi/2$. Thus the desired mapping is

$$s = 4\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right)$$

The resulting digital filter has the system function

$$H(z) = \frac{0.128 + 0.006z^{-1} - 0.122z^{-2}}{1 + 0.0006z^{-1} + 0.975z^{-2}}$$

We note that the coefficient of the z^{-1} term in the denominator of $H(z)$ is extremely small and can be approximated by zero. Thus we have the system function

$$H(z) = \frac{0.128 + 0.006z^{-1} - 0.122z^{-2}}{1 + 0.975z^{-2}}$$

This filter has poles at

$$p_{1,2} = 0.987e^{\pm j\pi/2}$$

and zeros at

$$z_{1,2} = -1, 0.95$$

Therefore, we have succeeded in designing a two-pole filter that resonates near $\omega = \pi/2$.

In this example the parameter T was selected to map the resonant frequency of the analog filter into the desired resonant frequency of the digital filter. Usually, the design of the digital filter begins with specifications in the digital domain, which involve the frequency variable ω. These specifications in frequency are converted to the analog domain by means of the relation in (8.3.43). The analog filter is then designed that meets these specifications and converted to a digital filter by means of the bilinear transformation in (8.3.40). In this procedure, the parameter T is transparent and may be set to any arbitrary value (e.g., $T = 1$). The following example illustrates this point.

Example 8.3.5

Design a single-pole lowpass digital filter with a 3-dB bandwidth of 0.2π, using the bilinear transformation applied to the analog filter

$$H(s) = \frac{\Omega_c}{s + \Omega_c}$$

where Ω_c is the 3-dB bandwidth of the analog filter.

Solution The digital filter is specified to have its -3-dB gain at $\omega_c = 0.2\pi$. In the frequency domain of the analog filter $\omega_c = 0.2\pi$ corresponds to

$$\begin{aligned}\Omega_c &= \frac{2}{T}\tan 0.1\pi \\ &= \frac{0.65}{T}\end{aligned}$$

Thus the analog filter has the system function

$$H(s) = \frac{0.65/T}{s + 0.65/T}$$

This represents our filter design in the analog domain.

Now, we apply the bilinear transformation given by (8.3.40) to convert the analog filter into the desired digital filter. Thus we obtain

$$H(z) = \frac{0.245(1 + z^{-1})}{1 - 0.509z^{-1}}$$

where the parameter T has been divided out.

The frequency response of the digital filter is

$$H(\omega) = \frac{0.245(1 + e^{-j\omega})}{1 - 0.509e^{-j\omega}}$$

At $\omega = 0$, $H(0) = 1$, and at $\omega = 0.2\pi$, we have $|H(0.2\pi)| = 0.707$, which is the desired response.

8.3.4 The Matched-z Transformation

Another method for converting an analog filter into an equivalent digital filter is to map the poles and zeros of $H(s)$ directly into poles and zeros in the z-plane. Suppose that the system function of the analog filter is expressed in the factored form

$$H(s) = \frac{\prod_{k=1}^{M}(s - z_k)}{\prod_{k=1}^{N}(s - p_k)} \tag{8.3.45}$$

where $\{z_k\}$ are the zeros and $\{p_k\}$ are the poles of the filter. Then the system function for the digital filter is

$$H(z) = \frac{\prod_{k=1}^{M}(1 - e^{z_k T} z^{-1})}{\prod_{k=1}^{N}(1 - e^{p_k T} z^{-1})} \tag{8.3.46}$$

where T is the sampling interval. Thus each factor of the form $(s - a)$ in $H(s)$ is mapped into the factor $(1 - e^{aT} z^{-1})$. This mapping is called the *matched-z transformation*.

We observe that the poles obtained from the matched-z transformation are identical to the poles obtained with the impulse invariance method. However, the two techniques result in different zero positions.

To preserve the frequency response characteristic of the analog filter, the sampling interval in the matched-z transformation must be properly selected to yield the pole and zero locations at the equivalent position in the z-plane. Thus aliasing must be avoided by selecting T sufficiently small.

8.3.5 Characteristics of Commonly Used Analog Filters

As we have seen from our discussion above, IIR digital filters can easily be obtained by beginning with an analog filter and then using a mapping to transform the s-plane into the z-plane. Thus the design of a digital filter is reduced to designing an appropriate analog filter and then performing the conversion from $H(s)$ to $H(z)$, in such a way so as to preserve as much as possible, the desired characteristics of the analog filter.

Analog filter design is a well-developed field and many books have been written on the subject. In this section we briefly describe the important characteristics of commonly used analog filters and introduce the relevant filter parameters. Our discussion is limited to lowpass filters. Subsequently, we describe several frequency transformations that convert a lowpass prototype filter into either a bandpass, highpass, or band-elimination filter.

Butterworth filters. Lowpass Butterworth filters are all-pole filters characterized by the magnitude-squared frequency response

$$|H(\Omega)|^2 = \frac{1}{1+(\Omega/\Omega_c)^{2N}} = \frac{1}{1+\epsilon^2(\Omega/\Omega_p)^{2N}} \tag{8.3.47}$$

where N is the order of the filter, Ω_c is its -3-dB frequency (usually called the cutoff frequency), Ω_p is the passband edge frequency, and $1/(1+\epsilon^2)$ is the band-edge value of $|H(\Omega)|^2$. Since $H(s)H(-s)$ evaluated at $s = j\Omega$ is simply equal to $|H(\Omega)|^2$, it follows that

$$H(s)H(-s) = \frac{1}{1+(-s^2/\Omega_c^2)^N} \tag{8.3.48}$$

The poles of $H(s)H(-s)$ occur on a circle of radius Ω_c at equally spaced points. From (8.3.48) we find that

$$\frac{-s^2}{\Omega_c^2} = (-1)^{1/N} = e^{j(2k+1)\pi/N} \qquad k = 0, 1, \ldots, N-1$$

and hence

$$s_k = \Omega_c e^{j\pi/2} e^{j(2k+1)\pi/2N} \qquad k = 0, 1, \ldots, N-1 \tag{8.3.49}$$

For example, Fig. 8.37 illustrates the pole positions for an $N = 4$ and $N = 5$ Butterworth filters.

The frequency response characteristics of the class of Butterworth filters are shown in Fig. 8.38 for several values of N. We note that $|H(\Omega)|^2$ is monotonic in both the passband and stopband. The order of the filter required to meet an attenuation δ_2 at a specified frequency Ω_s is easily determined from (8.3.47). Thus at $\Omega = \Omega_s$ we have

$$\frac{1}{1+\epsilon^2(\Omega_s/\Omega_p)^{2N}} = \delta_2^2$$

and hence

$$N = \frac{\log[(1/\delta_2^2) - 1]}{2\log(\Omega_s/\Omega_c)} = \frac{\log(\delta/\epsilon)}{\log(\Omega_s/\Omega_p)} \tag{8.3.50}$$

where, by definition, $\delta_2 = 1/\sqrt{1+\delta^2}$. Thus the Butterworth filter is completely characterized by the parameters N, δ_2, ϵ, and the ratio Ω_s/Ω_p.

Example 8.3.6

Determine the order and the poles of a lowpass Butterworth filter that has a -3-dB bandwidth of 500 Hz and an attenuation of 40 dB at 1000 Hz.

Solution The critical frequencies are the -3-dB frequency Ω_c and the stopband frequency Ω_s, which are

$$\Omega_c = 1000\pi$$

$$\Omega_s = 2000\pi$$

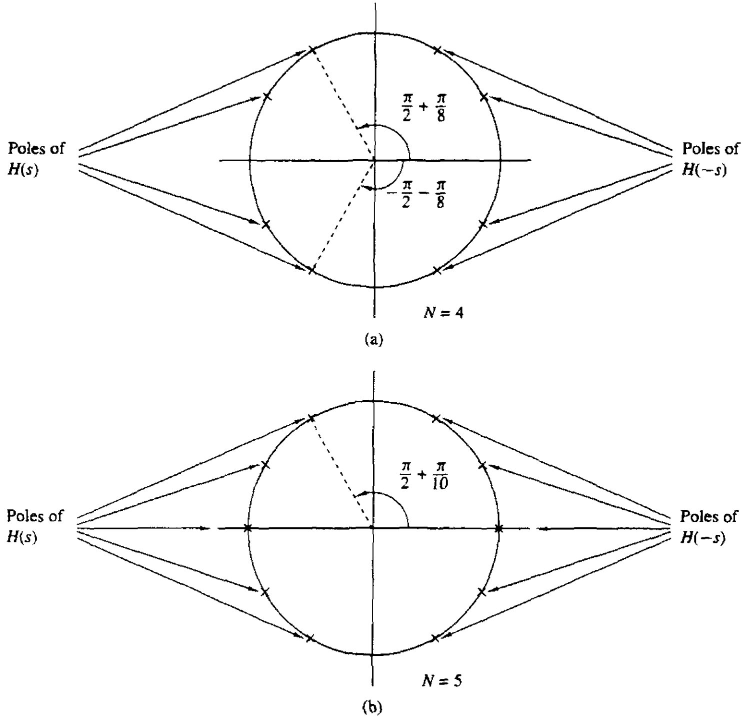

Figure 8.37 Pole positions for Butterworth filters.

For an attenuation of 40 dB, $\delta_2 = 0.01$. Hence from (8.3.50) we obtain

$$N = \frac{\log_{10}(10^4 - 1)}{2\log_{10} 2}$$
$$= 6.64$$

To meet the desired specifications, we select $N = 7$. The pole positions are

$$s_k = 1000\pi e^{j[\pi/2+(2k+1)\pi/14]} \qquad k = 0, 1, 2, \ldots, 6$$

Chebyshev filters. There are two types of Chebyshev filters. Type I Chebyshev filters are all-pole filters that exhibit equiripple behavior in the passband and a monotonic characteristic in the stopband. On the other hand, the family of type II Chebyshev filters contains both poles and zeros and exhibits a

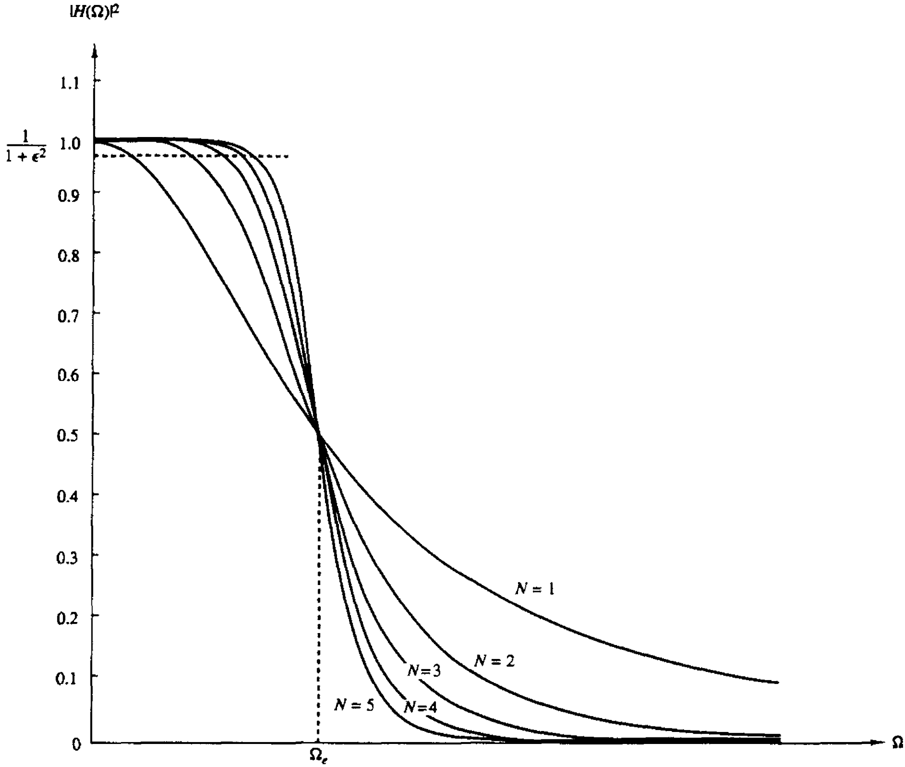

Figure 8.38 Frequency response of Butterworth filters.

monotonic behavior in the passband and an equiripple behavior in the stopband. The zeros of this class of filters lie on the imaginary axis in the s-plane.

The magnitude squared of the frequency response characteristic of a type I Chebyshev filter is given as

$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2 T_N^2(\Omega/\Omega_p)} \tag{8.3.51}$$

where ϵ is a parameter of the filter related to the ripple in the passband and $T_N(x)$ is the Nth-order Chebyshev polynomial defined as

$$T_N(x) = \begin{cases} \cos(N \cos^{-1} x), & |x| \le 1 \\ \cosh(N \cosh^{-1} x), & |x| > 1 \end{cases} \tag{8.3.52}$$

The Chebyshev polynomials can be generated by the recursive equation

$$T_{N+1}(x) = 2xT_N(x) - T_{N-1}(x) \qquad N = 1, 2, \ldots \tag{8.3.53}$$

where $T_0(x) = 1$ and $T_1(x) = x$. From (8.3.53) we obtain $T_2(x) = 2x^2 - 1$, $T_3(x) = 4x^3 - 3x$, and so on.

Some of the properties of these polynomials are as follows:

1. $|T_N(x)| \le 1$ for all $|x| \le 1$.
2. $T_N(1) = 1$ for all N.
3. All the roots of the polynomial $T_N(x)$ occur in the interval $-1 \le x \le 1$.

The filter parameter ϵ is related to the ripple in the passband, as illustrated in Fig. 8.39, for N odd and N even. For N odd, $T_N(0) = 0$ and hence $|H(0)|^2 = 1$. On the other hand, for N even, $T_N(0) = 1$ and hence $|H(0)|^2 = 1/(1+\epsilon^2)$. At the band edge frequency $\Omega = \Omega_p$, we have $T_N(1) = 1$, so that

$$\frac{1}{\sqrt{1+\epsilon^2}} = 1 - \delta_1$$

or, equivalently,

$$\epsilon^2 = \frac{1}{(1-\delta_1)^2} - 1 \tag{8.3.54}$$

where δ_1 is the value of the passband ripple.

The poles of a type I Chebyshev filter lie on an ellipse in the s-plane with major axis

$$r_1 = \Omega_p \frac{\beta^2 + 1}{2\beta} \tag{8.3.55}$$

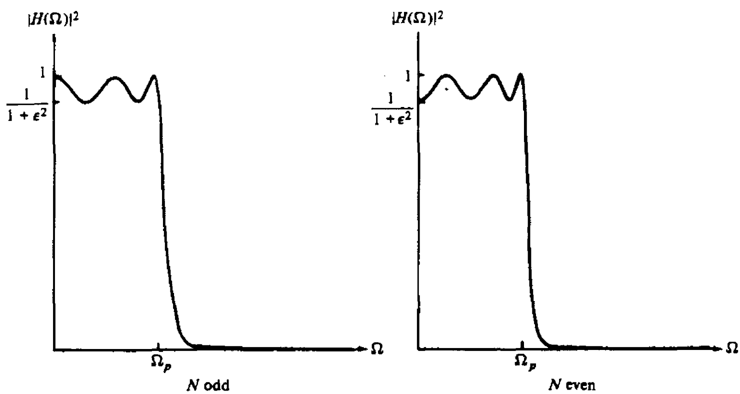

Figure 8.39 Type I Chebyshev filter characteristic.

and minor axis

$$r_2 = \Omega_p \frac{\beta^2 - 1}{2\beta} \tag{8.3.56}$$

where β is related to ϵ according to the equation

$$\beta = \left[\frac{\sqrt{1+\epsilon^2}+1}{\epsilon}\right]^{1/N} \tag{8.3.57}$$

The pole locations are most easily determined for a filter of order N by first locating the poles for an equivalent Nth-order Butterworth filter that lie on circles of radius r_1 or radius r_2, as illustrated in Fig. 8.40. If we denote the angular positions of the poles of the Butterworth filter as

$$\phi_k = \frac{\pi}{2} + \frac{(2k+1)\pi}{2N} \qquad k = 0, 1, 2, \ldots, N-1 \tag{8.3.58}$$

then the positions of the poles for the Chebyshev filter lie on the ellipse at the coordinates (x_k, y_k), $k = 0, 1, \ldots, N-1$, where

$$\begin{aligned} x_k &= r_2 \cos\phi_k, \qquad k = 0, 1, \ldots, N-1 \\ y_k &= r_1 \sin\phi_k, \qquad k = 0, 1, \ldots, N-1 \end{aligned} \tag{8.3.59}$$

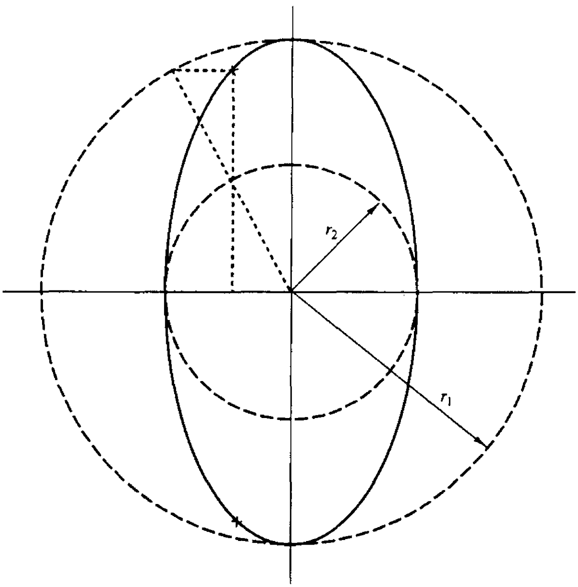

Figure 8.40 Determination of the pole locations for a Chebyshev filter.

A type II Chebyshev filter contains zeros as well as poles. The magnitude squared of its frequency response is given as

$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2[T_N^2(\Omega_s/\Omega_p)/T_N^2(\Omega_s/\Omega)]} \tag{8.3.60}$$

where $T_N(x)$ is, again, the Nth-order Chebyshev polynomial and Ω_s is the stopband frequency as illustrated in Fig. 8.41. The zeros are located on the imaginary axis at the points

$$s_k = j\frac{\Omega_s}{\sin\phi_k} \qquad k = 0, 1, \ldots, N-1 \tag{8.3.61}$$

The poles are located at the points (v_k, w_k), where

$$v_k = \frac{\Omega_s x_k}{\sqrt{x_k^2 + y_k^2}} \qquad k = 0, 1, \ldots, N-1 \tag{8.3.62}$$

$$w_k = \frac{\Omega_s y_k}{\sqrt{x_k^2 + y_k^2}} \qquad k = 0, 1, \ldots, N-1 \tag{8.3.63}$$

where $\{x_k\}$ and $\{y_k\}$ are defined in (8.3.59) with β now related to the ripple in the stopband through the equation

$$\beta = \left[\frac{1 + \sqrt{1 - \delta_2^2}}{\delta_2}\right]^{1/N} \tag{8.3.64}$$

From this description, we observe that the Chebyshev filters are characterized by the parameters N, ϵ, δ_2, and the ratio Ω_s/Ω_p. For a given set of specifications

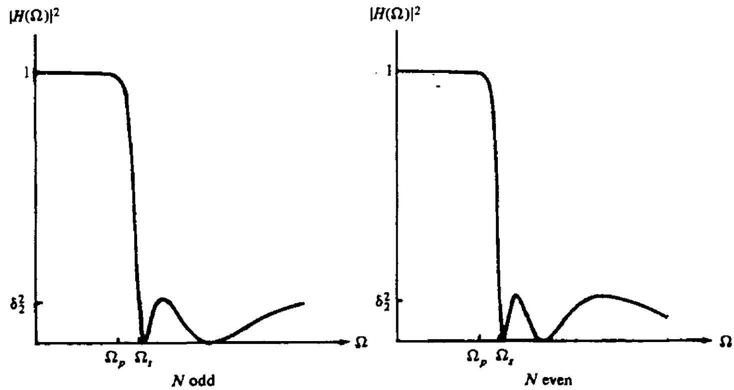

Figure 8.41 Type II Chebyshev filters.

on ϵ, δ_2, and Ω_s/Ω_p, we can determine the order of the filter from the equation

$$N = \frac{\log\left[\left(\sqrt{1-\delta_2^2}+\sqrt{1-\delta_2^2(1+\epsilon^2)}\right)/\epsilon\delta_2\right]}{\log\left[(\Omega_s/\Omega_p)+\sqrt{(\Omega_s/\Omega_p)^2-1}\right]} \tag{8.3.65}$$

$$= \frac{\cosh^{-1}(\delta/\epsilon)}{\cosh^{-1}(\Omega_s/\Omega_p)}$$

where, by definition, $\delta_2 = 1/\sqrt{1+\delta^2}$.

Example 8.3.7

Determine the order and the poles of a type I lowpass Chebyshev filter that has a 1-dB ripple in the passband, a cutoff frequency $\Omega_p = 1000\pi$, a stopband frequency of 2000π, and an attenuation of 40 dB or more for $\Omega \geq \Omega_s$.

Solution First, we determine the order of the filter. We have

$$\begin{aligned} 10\log_{10}(1+\epsilon^2) &= 1 \\ 1+\epsilon^2 &= 1.259 \\ \epsilon^2 &= 0.259 \\ \epsilon &= 0.5088 \end{aligned}$$

Also,

$$\begin{aligned} 20\log_{10}\delta_2 &= -40 \\ \delta_2 &= 0.01 \end{aligned}$$

Hence from (8.3.65) we obtain

$$\begin{aligned} N &= \frac{\log_{10}196.54}{\log_{10}(2+\sqrt{3})} \\ &= 4.0 \end{aligned}$$

Thus a type I Chebyshev filter having four poles meets the specifications.

The pole positions are determined from the relations in (8.3.55) through (8.3.59). First, we compute β, r_1, and r_2. Hence

$$\begin{aligned} \beta &= 1.429 \\ r_1 &= 1.06\Omega_p \\ r_2 &= 0.365\Omega_p \end{aligned}$$

The angles $\{\phi_k\}$ are

$$\phi_k = \frac{\pi}{2} + \frac{(2k+1)\pi}{8} \qquad k = 0, 1, 2, 3$$

Therefore, the poles are located at

$$\begin{aligned} x_1 + jy_1 &= -0.1397\Omega_p \pm j0.979\Omega_p \\ x_2 + jy_2 &= -0.337\Omega_p \pm j0.4056\Omega_p \end{aligned}$$

The filter specifications in Example 8.3.7 are very similar to the specifications given in Example 8.3.6, which involved the design of a Butterworth filter. In that case the number of poles required to meet the specifications was seven. On the other hand, the Chebyshev filter required only four. This result is typical of such comparisons. In general, the Chebyshev filter meets the specifications with a fewer number of poles than the corresponding Butterworth filter. Alternatively, if we compare a Butterworth filter to a Chebyshev filter having the same number of poles and the same passband and stopband specifications, the Chebyshev filter will have a smaller transition bandwidth. For a tabulation of the characteristics of Chebyshev filters and their pole–zero locations, the interested reader is referred to the handbook of Zverev (1967).

Elliptic filters. Elliptic (or Cauer) filters exhibit equiripple behavior in both the passband and the stopband, as illustrated in Fig. 8.42 for N odd and N even. This class of filters contains both poles and zeros and is characterized by the magnitude-squared frequency response

$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2 U_N(\Omega/\Omega_p)} \tag{8.3.66}$$

where $U_N(x)$ is the Jacobian elliptic function of order N, which has been tabulated by Zverev (1967), and ϵ is a parameter related to the passband ripple. The zeros lie on the $j\Omega$-axis.

We recall from our discussion of FIR filters that the most efficient designs occur when we spread the approximation error equally over the passband and the stopband. Elliptic filters accomplish this objective and, as a consequence, are the most efficient from the viewpoint of yielding the smallest-order filter for a given

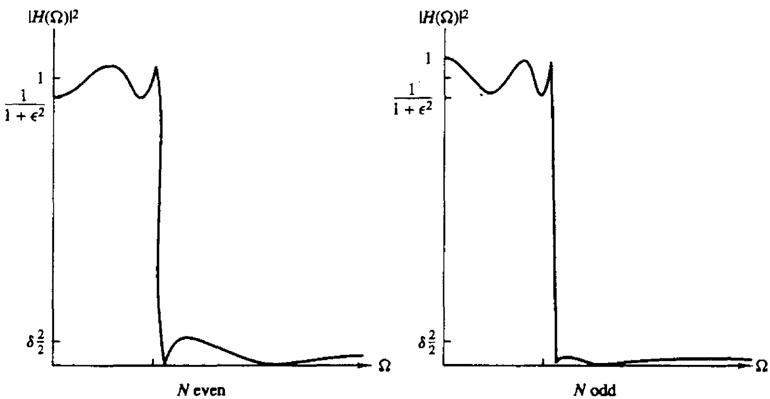

Figure 8.42 Magnitude-squared frequency characteristics of elliptic filters.

set of specifications. Equivalently, we can say that for a given order and a given set of specifications, an elliptic filter has the smallest transition bandwidth.

The filter order required to achieve a given set of specifications in passband ripple δ_1, stopband ripple δ_2, and transition ratio Ω_p/Ω_s is given as

$$N = \frac{K(\Omega_p/\Omega_s)K\left(\sqrt{1-(\epsilon^2/\delta^2)}\right)}{K(\epsilon/\delta)K\left(\sqrt{1-(\Omega_p/\Omega_s)^2}\right)} \tag{8.3.67}$$

where $K(x)$ is the complete elliptic integral of the first kind, defined as

$$K(x) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1-x^2\sin^2\theta}} \tag{8.3.68}$$

and $\delta_2 = 1/\sqrt{1+\delta^2}$. Values of this integral have been tabulated in a number of texts [e.g., the books by Jahnke and Emde (1945) and Dwight (1957)]. The passband ripple is $10\log_{10}(1+\epsilon^2)$.

We shall not attempt to describe elliptic functions in any detail because such a discussion would take us too far afield. Suffice to say that computer programs are available for designing elliptic filters from the frequency specifications indicated above.

In view of the optimality of elliptic filters, the reader may question the reason for considering the class of Butterworth or the class of Chebyshev filters in practical applications. One important reason that these other types of filters might be preferable in some applications is that they possess better phase response characteristics. The phase response of elliptic filters is more nonlinear in the passband than a comparable Butterworth filter or a Chebyshev filter, especially near the band edge.

Bessel filters. Bessel filters are a class of all-pole filters that are characterized by the system function

$$H(s) = \frac{1}{B_N(s)} \tag{8.3.69}$$

where $B_N(s)$ is the Nth-order Bessel polynomial. These polynomials can be expressed in the form

$$B_N(s) = \sum_{k=0}^{N} a_k s^k \tag{8.3.70}$$

where the coefficients $\{a_k\}$ are given as

$$a_k = \frac{(2N-k)!}{2^{N-k}k!(N-k)!} \qquad k = 0, 1, \ldots, N \tag{8.3.71}$$

Alternatively, the Bessel polynomials may be generated recursively from the relation

$$B_N(s) = (2N-1)B_{N-1}(s) + s^2 B_{N-2}(s) \tag{8.3.72}$$

with $B_0(s) = 1$ and $B_1(s) = s + 1$ as initial conditions.

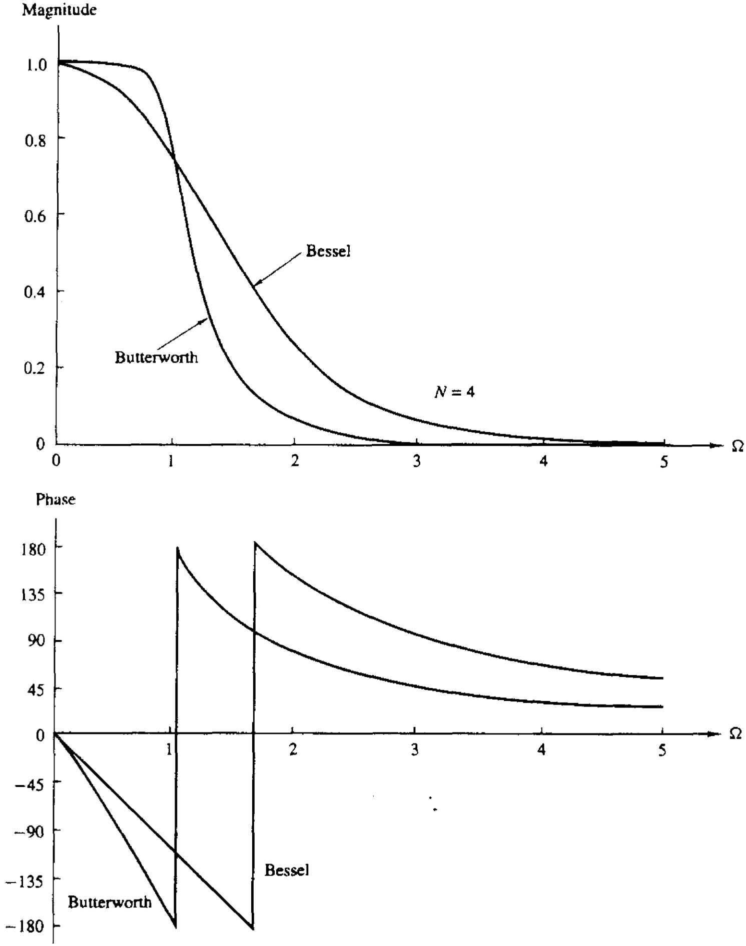

Figure 8.43 Magnitude and phase responses of Bessel and Butterworth filters of order $N = 4$.

An important characteristic of Bessel filters is the linear-phase response over the passband of the filter. For example, Fig. 8.43 shows a comparison of the magnitude and phase responses of a Bessel filter and Butterworth filter of order $N = 4$. We note that the Bessel filter has a larger transition bandwidth, but its phase is linear within the passband. However, we should emphasize that the linear-phase chacteristics of the analog filter are destroyed in the process of converting the filter into the digital domain by means of the transformations decribed previously.

8.3.6 Some Examples of Digital Filter Designs Based on the Bilinear Transformation

In this section we present several examples of digital filter designs obtained from analog filters by applying the bilinear transformation to convert $H(s)$ to $H(z)$. These filters designs are performed with the aid of one of several software packages now available for use on a personal computer.

A lowpass filter is designed to meet specifications of a maximum ripple of $\frac{1}{2}$ dB in the passband, 60-dB attenuation in the stopband, a passband edge frequency of $\omega_p = 0.25\pi$, and a stopband edge frequency of $\omega_s = 0.30\pi$.

A Butterworth filter of order $N = 37$ is required to satisfy the specifications. Its frequency response characteristics are illustrated in Fig. 8.44. If a Chebyshev filter is used, a filter of order $N = 13$ satisfies the specifications. The frequency response characteristics for a type I and type II Chebyshev filters are shown in Figs. 8.45 and 8.46, respectively. The type I filter has a passband ripple of 0.31 dB. Finally, an elliptic filter of order $N = 7$ is designed which also satisfied the specifications. For illustrative purposes, we show in Table 8.11, the numerical values for the filter parameters and the resulting frequency specifications are shown in Fig. 8.47. The following notation is used for the parameters in the function $H(z)$:

$$H(z) = \prod_{i=1}^{K} \frac{b(i,0) + b(i,1)z^{-1} + b(i,2)z^{-2}}{1 + a(i,1)z^{-1} + a(i,2)z^{-2}} \tag{8.3.73}$$

Although we have described only lowpass analog filters in the preceding section, it is a simple matter to convert a lowpass analog filter into a bandpass, bandstop, or highpass analog filter by a frequency transformation, as is described in Section 8.4. The bilinear transformation is then applied to convert the analog filter into an equivalent digital filter. As in the case of the lowpass filters described above, the entire design can be carried out on a computer.

8.4 FREQUENCY TRANSFORMATIONS

The treatment in the preceding section is focused primarily on the design of lowpass IIR filters. If we wish to design a highpass or a bandpass or a bandstop filter, it is a simple matter to take a lowpass prototype filter (Butterworth, Chebyshev, elliptic, Bessel) and perform a frequency transformation.

One possibility is to perform the frequency transformation in the analog domain and then to convert the analog filter into a corresponding digital filter by a mapping of the s-plane into the z-plane. An alternative approach is first to

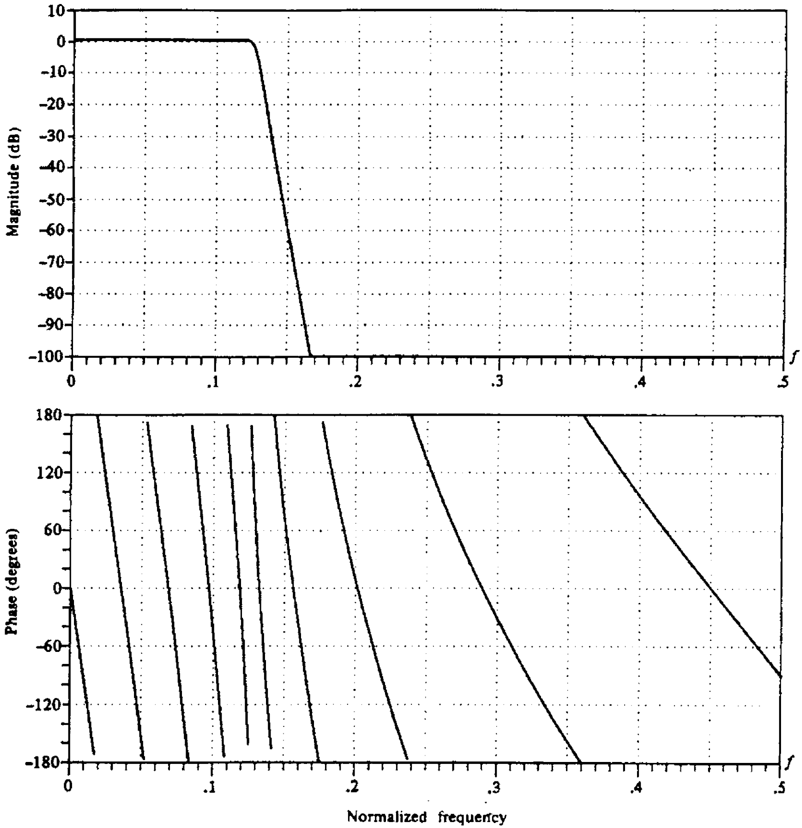

Figure 8.44 Frequency response characteristics of a 37-order Butterworth filter.

convert the analog lowpass filter into a lowpass digital filter and then to transform the lowpass digital filter into the desired digital filter by a digital transformation. In general, these two approaches yield different results, except for the bilinear transformation, in which case the resulting filter designs are identical. These two approaches are described below.

8.4.1 Frequency Transformations in the Analog Domain

First, we consider frequency transformations in the analog domain. Suppose that we have a lowpass filter with passband edge frequency Ω_p and we wish to convert

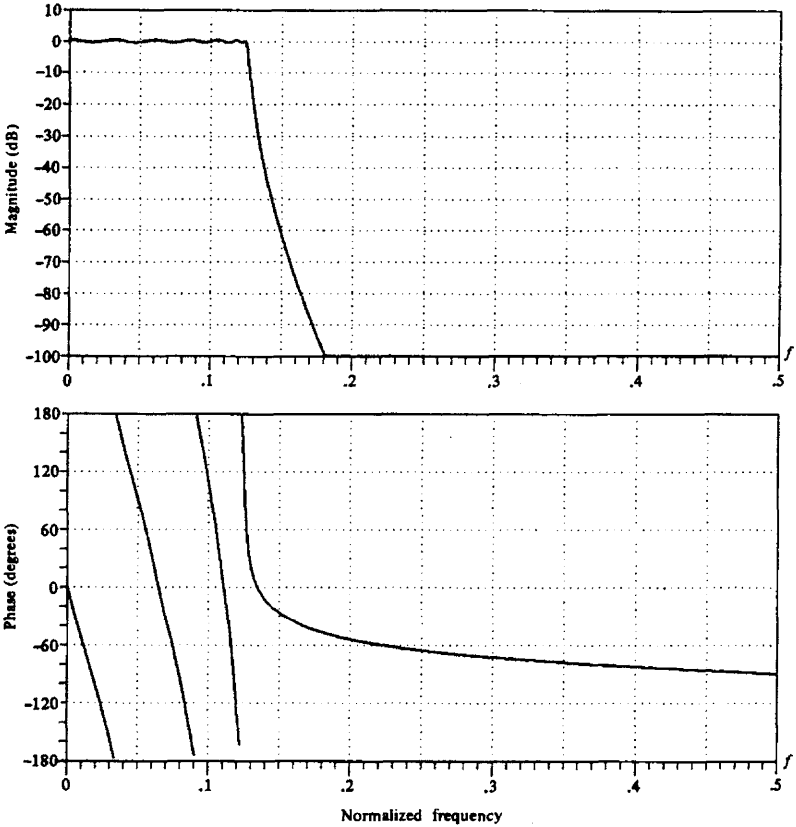

Figure 8.45 Frequency response characteristics of a 13-order type I Chebyshev filter.

it to another lowpass filter with passband edge frequency Ω'_p. The transformation that accomplishes this is

$$s \longrightarrow \frac{\Omega_p}{\Omega'_p}s \qquad \text{(lowpass to lowpass)} \tag{8.4.1}$$

Thus we obtain a lowpass filter with system function $H_l(s) = H_p[(\Omega_p/\Omega'_p)s]$, where $H_p(s)$ is the system function of the prototype filter with passband edge frequency Ω_p.

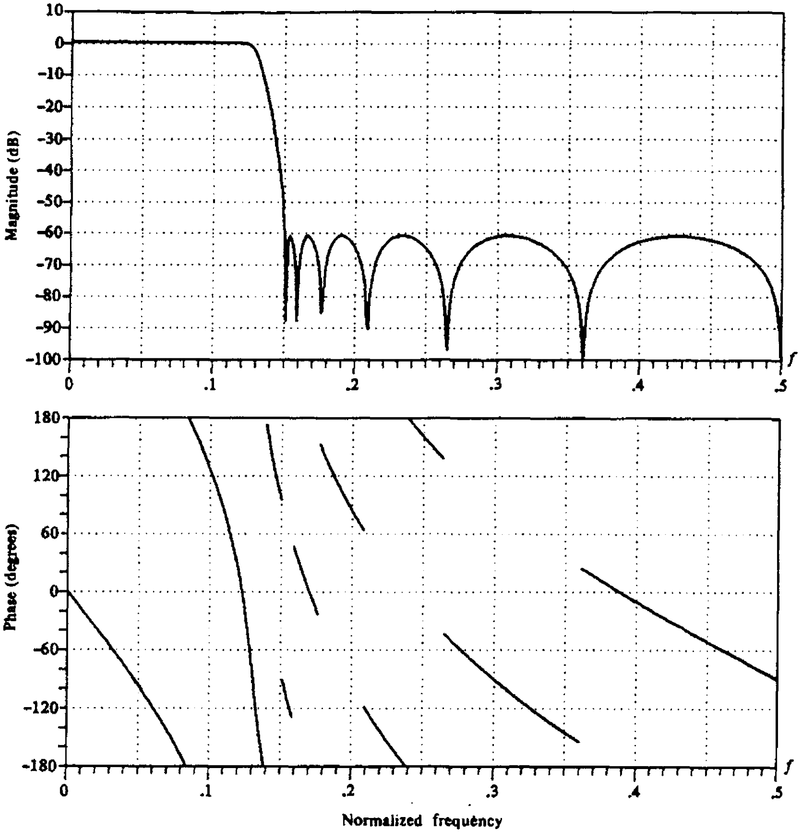

Figure 8.46 Frequency response characteristics of a 13-order type II Chebyshev filter.

If we wish to convert a lowpass filter into a highpass filter with passband edge frequency Ω'_p, the desired transformation is

$$s \longrightarrow \frac{\Omega_p \Omega'_p}{s} \qquad \text{(lowpass to highpass)} \tag{8.4.2}$$

The system function of the highpass filter is $H_h(s) = H_p(\Omega_p \Omega'_p / s)$.

The transformation for converting a lowpass analog filter with passband edge frequency Ω_p into a band filter, having a lower band edge frequency Ω_l and an upper band edge frequency Ω_u, can be accomplished by first converting the lowpass

TABLE 8.11 FILTER COEFFICIENTS FOR A 7-ORDER ELLIPTIC FILTER

INFINITE IMPULSE RESPONSE (IIR)
ELLIPTIC LOWPASS FILTER
UNQUANTIZED COEFFICIENTS
FILTER ORDER = 7
SAMPLING FREQUENCY = 2.000 KILOHERTZ

I.	A(I, 1)	A(I, 2)	B(I, 0)	B(I, 1)	B(I, 2)
1	-.790103	.000000	.104948	.104948	.000000
2	-1.517223	.714088	.102450	-.007817	.102232
3	-1.421773	.861895	.420100	-.399842	.419864
4	-1.387447	.962252	.714929	-.826743	.714841

*** CHARACTERISTICS OF DESIGNED FILTER ***

	BAND 1	BAND 2
LOWER BAND EDGE	.00000	.30000
UPPER BAND EDGE	.25000	1.00000
NOMINAL GAIN	1.00000	.00000
NOMINAL RIPPLE	.05600	.00100
MAXIMUM RIPPLE	.04910	.00071
RIPPLE IN DB	.41634	-63.00399

filter into another lowpass filter having a band edge frequency $\Omega'_p = 1$ and then performing the transformation

$$s \longrightarrow \frac{s^2 + \Omega_l \Omega_u}{s(\Omega_u - \Omega_l)} \qquad \text{(lowpass to bandpass)} \tag{8.4.3}$$

Equivalently, we can accomplish the same result in a single step by means of the transformation

$$s \longrightarrow \Omega_p \frac{s^2 + \Omega_l \Omega_u}{s(\Omega_u - \Omega_l)} \qquad \text{(lowpass to bandpass)} \tag{8.4.4}$$

where

$$\Omega_l = \text{lower band edge frequency}$$

$$\Omega_u = \text{upper band edge frequency}$$

Thus we obtain

$$H_b(s) = H_p\left(\Omega_p \frac{s^2 + \Omega_l \Omega_u}{s(\Omega_u - \Omega_l)}\right)$$

Finally, if we wish to convert a lowpass analog filter with band edge frequency Ω_p into a bandstop filter, the transformation is simply the inverse of (8.4.3) with the additional factor Ω_p serving to normalize for the band edge frequency of the lowpass filter. Thus the transformation is

$$s \longrightarrow \Omega_p \frac{s(\Omega_u - \Omega_l)}{s^2 + \Omega_u \Omega_l} \qquad \text{(lowpass to bandstop)} \tag{8.4.5}$$

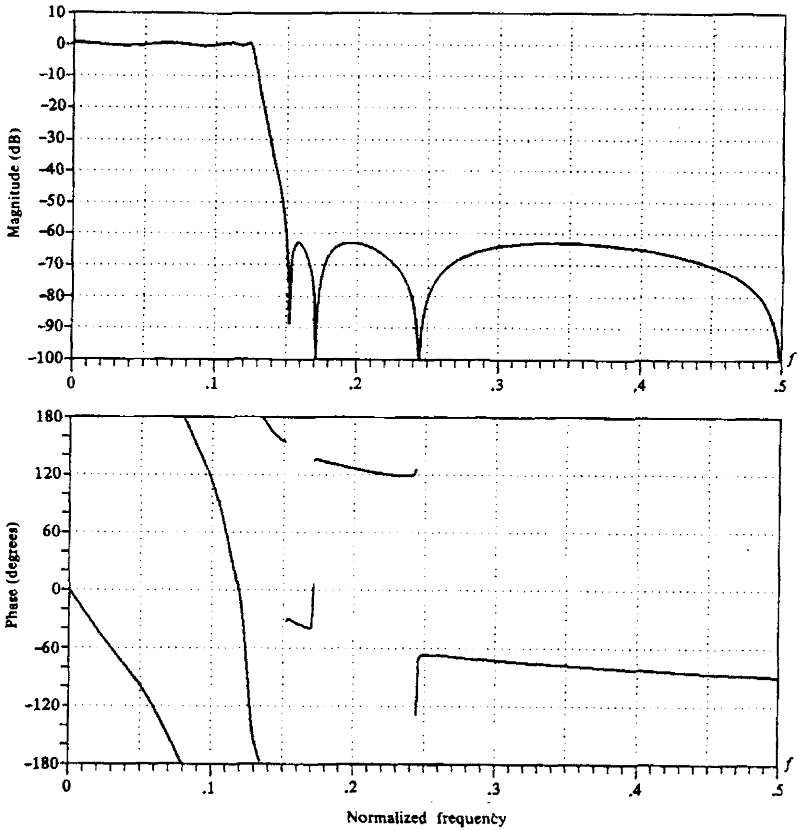

Figure 8.47 Frequency response characteristics of a 7-order elliptic filter.

which leads to

$$H_{bs}(s) = H_p\left(\Omega_p \frac{s(\Omega_u - \Omega_l)}{s^2 + \Omega_u \Omega_l}\right)$$

The mappings in (8.4.1), (8.4.2), (8.4.3), and (8.4.5) are summarized in Table 8.12. The mappings in (8.4.4) and (8.4.5) are nonlinear and may appear to distort the frequency response characteristics of the lowpass filter. However, the effects of the nonlinearity on the frequency response are minor, primarily affecting the frequency scale but preserving the amplitude response characteristics of the filter. Thus an equiripple lowpass filter is transformed into an equiripple bandpass or bandstop or highpass filter.

TABLE 8.12 FREQUENCY TRANSFORMATIONS FOR ANALOG FILTERS (PROTOTYPE LOWPASS FILTER HAS BAND EDGE FREQUENCY Ω_P)

Type of transformation	Transformation	Band edge frequencies of new filter
Lowpass	$s \longrightarrow \dfrac{\Omega_p}{\Omega'_p} s$	Ω'_p
Highpass	$s \longrightarrow \dfrac{\Omega_p \Omega'_p}{s}$	Ω'_p
Bandpass	$s \longrightarrow \Omega_p \dfrac{s^2 + \Omega_l \Omega_u}{s(\Omega_u - \Omega_l)}$	Ω_l, Ω_u
Bandstop	$s \longrightarrow \Omega_p \dfrac{s(\Omega_u - \Omega_c)}{s^2 + \Omega_u \Omega_l}$	Ω_l, Ω_u

Example 8.4.1

Transform the single-pole lowpass Butterworth filter with system function

$$H(s) = \frac{\Omega_p}{s + \Omega_p}$$

into a bandpass filter with upper and lower band edge frequencies Ω_u and Ω_l, respectively.

Solution The desired transformation is given by (8.3.4). Thus we have

$$\begin{aligned} H(s) &= \frac{1}{\dfrac{s^2 + \Omega_l \Omega_u}{s(\Omega_u - \Omega_l)} + 1} \\ &= \frac{(\Omega_u - \Omega_l)s}{s^2 + (\Omega_u - \Omega_l)s + \Omega_l \Omega_u} \end{aligned}$$

The resulting filter has a zero at s = 0 and poles at

$$s = \frac{-(\Omega_u - \Omega_l) \pm \sqrt{\Omega_u^2 + \Omega_l^2 - 6\Omega_u \Omega_l}}{2}$$

8.4.2 Frequency Transformations in the Digital Domain

As in the analog domain, frequency transformations can be performed on a digital lowpass filter to convert it to either a bandpass, bandstop, or highpass filter. The transformation involves replacing the variable z^{-1} by a rational function $g(z^{-1})$, which must satisfy the following properties.

1. The mapping $z^{-1} \longrightarrow g(z^{-1})$ must map points inside the unit circle in the z-plane into itself.
2. The unit circle must also be mapped into itself.

Condition (2) implies that for $r = 1$,

$$\begin{aligned} e^{-j\omega} &= g(e^{-j\omega}) \equiv g(\omega) \\ &= |g(\omega)|e^{j\arg[g(\omega)]} \end{aligned}$$

It is clear that we must have $|g(\omega)| = 1$ for all ω. That is, the mapping must be all-pass. Hence it is of the form

$$g(z^{-1}) = \pm \prod_{k=1}^{n} \frac{z^{-1} - a_k}{1 - a_k z^{-1}} \tag{8.4.6}$$

where $|a_k| < 1$ to ensure that a stable filter is transformed into another stable filter (i.e., to satisfy condition 1).

From the general form in (8.4.6), we obtain the desired set of digital transformations for converting a prototype digital lowpass filter into either a bandpass, a bandstop, a highpass, or another lowpass digital filter. These transformations are tabulated in Table 8.13.

TABLE 8.13 FREQUENCY TRANSFORMATION FOR DIGITAL FILTERS (PROTOTYPE LOWPASS FILTER HAS BAND EDGE FREQUENCY ω_p)

Type of transformation	Transformation	Parameters
Lowpass	$z^{-1} \longrightarrow \dfrac{z^{-1} - a}{1 - az^{-1}}$	ω'_p = band edge frequency of new filter $a = \dfrac{\sin[(\omega_p - \omega'_p)/2]}{\sin[(\omega_p + \omega'_p)/2]}$
Highpass	$z^{-1} \longrightarrow -\dfrac{z^{-1} + a}{1 + az^{-1}}$	ω'_p = band edge frequency new filter $a = -\dfrac{\cos[(\omega_p + \omega'_p)/2]}{\cos[(\omega_p - \omega'_p)/2]}$
Bandpass	$z^{-1} \longrightarrow -\dfrac{z^{-2} - a_1 z^{-1} + a_2}{a_2 z^{-2} - a_1 z^{-1} + 1}$	ω_l = lower band edge frequency ω_u = upper band edge frequency $a_1 = -2\alpha K/(K+1)$ $a_2 = (K-1)/(K+1)$ $\alpha = \dfrac{\cos[(\omega_u + \omega_l)/2]}{\cos[(\omega_u - \omega_l)/2]}$ $K = \cot\dfrac{\omega_u - \omega_l}{2}\tan\dfrac{\omega_p}{2}$
Bandstop	$z^{-1} \longrightarrow \dfrac{z^{-2} - a_1 z^{-1} + a_2}{a_2 z^{-2} - a_1 z^{-1} + 1}$	ω_l = lower band edge frequency ω_u = upper band edge frequency $a_1 = -2\alpha/(K+1)$ $a_2 = (1-K)/(1+K)$ $\alpha = \dfrac{\cos[(\omega_u + \omega_l)/2]}{\cos[(\omega_u - \omega_l)/2]}$ $K = \tan\dfrac{\omega_u - \omega_l}{2}\tan\dfrac{\omega_p}{2}$

Example 8.4.2

Convert the single-pole lowpass Butterworth filter with system function

$$H(z) = \frac{0.245(1 + z^{-1})}{1 - 0.509z^{-1}}$$

into a bandpass filter with upper and lower cutoff frequencies ω_u and ω_l, respectively. The lowpass filter has 3-dB bandwidth $\omega_p = 0.2\pi$ (see Example 8.3.5).

Solution The desired transformation is

$$z^{-1} \longrightarrow -\frac{z^{-2} - a_1 z^{-1} + a_2}{a_2 z^{-2} - a_1 z^{-1} + 1}$$

where a_1 and a_2 are defined in Table 8.13. Substitution into $H(z)$ yields

$$\begin{aligned} H(z) &= \frac{0.245\left[1 - \dfrac{z^{-2} - a_1 z^{-1} + a_2}{a_2 z^{-2} - a_1 z^{-1} + 1}\right]}{1 + 0.509\left(\dfrac{z^{-2} - a_1 z^{-1} + a_2}{a_2 z^{-2} - a_1 z^{-1} + 1}\right)} \\ &= \frac{0.245(1 - a_2)(1 - z^{-2})}{(1 + 0.509a_2) - 1.509a_1 z^{-1} + (a_2 + 0.509)z^{-2}} \end{aligned}$$

Note that the resulting filter has zeros at $z = \pm 1$ and a pair of poles that depend on the choice of ω_u and ω_l.

For example, suppose that $\omega_u = 3\pi/5$ and $\omega_l = 2\pi/5$. Since $\omega_p = 0.2\pi$, we find that $K = 1$, $a_2 = 0$, and $a_1 = 0$. Then

$$H(z) = \frac{0.245(1 - z^{-2})}{1 + 0.509z^{-2}}$$

This filter has poles at $z = \pm j0.713$ and hence resonates at $\omega = \pi/2$.

Since a frequency transformation can be performed either in the analog domain or in the digital domain, the filter designer has a choice as to which approach to take. However, some caution must be exercised depending on the types of filters being designed. In particular, we know that the impulse invariance method and the mapping of derivatives are inappropriate to use in designing highpass and many bandpass filters, due to the aliasing problem. Consequently, one would not employ an analog frequency transformation followed by conversion of the result into the digital domain by use of these two mappings. Instead, it is much better to perform the mapping from an analog lowpass filter into a digital lowpass filter by either of these mappings, and then to perform the frequency transformation in the digital domain. Thus the problem of aliasing is avoided.

In the case of the bilinear transformation, where aliasing is not a problem, it does not matter whether the frequency transformation is performed in the analog domain or in the digital domain. In fact, in this case only, the two approaches result in identical digital filters.

8.5 DESIGN OF DIGITAL FILTERS BASED ON LEAST-SQUARES METHOD

Except for the impulse invariance method, the design techniques for IIR filters described in Section 8.3 involved the conversion of an analog filter into a digital filter by some mapping from the s-plane to the z-plane. As an alternative, one can design digital IIR filters directly in the z-domain without reference to the analog domain.

We now describe several methods for designing digital filters directly. In the first three techniques, the Padé approximation method and least-squares design methods, the specifications are given in the time domain and the design is carried out in the time domain. The final section describes a least-squares technique in which the design is carried out in the frequency domain.

8.5.1 Padé Approximation Method

Suppose that the desired impulse response $h_d(n)$ is specified for $n \geq 0$. The filter to be designed has the system function

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}} = \sum_{k=0}^{\infty} h(k) z^{-k} \tag{8.5.1}$$

where $h(k)$ is its unit sample response. The filter has $L = M + N + 1$ parameters, namely, the coefficients $\{a_k\}$ and $\{b_k\}$, which can be selected to minimize some error criterion.

The least-squares error criterion is often used in optimization problems of this type. Suppose that we minimize the sum of the squared errors

$$\mathcal{E} = \sum_{n=0}^{U} [h_d(n) - h(n)]^2 \tag{8.5.2}$$

with respect to the filter parameters $\{a_k\}$ and $\{b_k\}$, where U is some preselected upper limit in the summation.

In general, $h(n)$ is a nonlinear function of the filter parameters and hence the minimization of $\mathcal{E}$ involves the solution of a set of nonlinear equations. However, if we select the upper limit as $U = L - 1$, it is possible to match $h(n)$ perfectly to the desired response $h_d(n)$ for $0 \leq n \leq M + N$. This can be achieved in the following manner.

The difference equation for the desired filter is

$$\begin{aligned} y(n) = &-a_1 y(n-1) - a_2 y(n-2) - \cdots - a_N y(n-N) \\ &+ b_0 x(n) + b_1 x(n-1) + \cdots + b_M x(n-M) \end{aligned} \tag{8.5.3}$$

Suppose that the input to the filter is a unit sample [i.e., $x(n) = \delta(n)$]. Then the response of the filter is $y(n) = h(n)$ and hence (8.5.3) becomes

$$h(n) = -a_1 h(n-1) - a_2 h(n-2) - \cdots - a_N h(n-N) + b_0\delta(n) + b_1\delta(n-1) + \cdots + b_M\delta(n-M) \tag{8.5.4}$$

Since $\delta(n-k) = 0$ except for $n = k$, (8.5.4) reduces to

$$h(n) = -a_1 h(n-1) - a_2 h(n-2) - \cdots - a_N h(n-N) + b_n \qquad 0 \le n \le M \tag{8.5.5}$$

For $n > M$, (8.5.4) becomes

$$h(n) = -a_1 h(n-1) - a_2 h(n-2) - \cdots - a_N h(n-N) \tag{8.5.6}$$

The set of linear equations in (8.5.5) and (8.5.6) can be used to solve for the filter parameters $\{a_k\}$ and $\{b_k\}$. We set $h(n) = h_d(n)$ for $0 \le n \le M + N$, and use the linear equations in (8.5.6) to solve for the filter parameters $\{a_k\}$. Then we use values for the $\{a_k\}$ in (8.5.5) and solve for the parameters $\{b_k\}$. Thus we obtain a perfect match between $h(n)$ and the desired response $h_d(n)$ for the first L values of the impulse response. This design technique is usually called the *Padé approximation procedure.*

The degree to which this design technique produces acceptable filter designs depends in part on the number of filter coefficients selected. Since the design method matches $h_d(n)$ only up to the number of filter parameters, the more complex the filter, the better the approximation to $h_d(n)$ for $0 \le n \le M+N$. However, this is also the major limitation with the Padé approximation method, namely, the resulting filter must contain a large number of poles and zeros. For this reason, the Padé approximation method has found limited use in filter designs for practical applications.

Example 8.5.1

Suppose that the desired unit sample response is

$$h_d(n) = 2(\tfrac{1}{2})^n u(n)$$

Determine the parameters of the filter with system function

$$H(z) = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1}}$$

using the Padé approximation technique.

Solution In this simple example, $H(z)$ can provide a perfect match to $H_d(z)$, by selecting $b_0 = 2$, $b_1 = 0$, and $a_1 = -\frac{1}{2}$. Let us apply the Padé approximation to see if we indeed obtain the same result.

With $\delta(n)$ as the input to $H(z)$, we obtain the output

$$h(n) = -a_1 h(n-1) + b_0\delta(n) + b_1\delta(n-1)$$

For $n > 1$, we have

$$h(n) = -a_1 h(n-1)$$

or, equivalently,

$$h_d(n) = -a_1 h_d(n-1)$$

With the substitution for $h_d(n)$, we obtain $a_1 = -\frac{1}{2}$. To solve for b_0 and b_1, we use the form (8.5.5) with $h(n) = h_d(n)$. Thus

$$h_d(n) = \tfrac{1}{2}h_d(n-1) + b_0\delta(n) + b_1\delta(n-1)$$

For $n = 0$ this equation yields $b_0 = 2$. For $n = 1$ we obtain the result $b_1 = 0$. Thus $H(z) = H_d(z)$.

This example illustrates that the Padé approximation results in a perfect match to $H_d(z)$ when the desired system function is rational and we have prior knowledge of the number of poles and zeros in the system. In general, however, this is not the case in practice, since $h_d(n)$ is determined from some desired frequency response specifications $H_d(\omega)$. In such a case the Padé approximation may not result in a good filter design. To illustrate a potential problem and suggest a solution, let us consider the following examples.

Example 8.5.2

A fourth-order Butterworth filter has the system function

$$H_d(z) = \frac{4.8334 \times 10^{-3}(z+1)^4}{(z^2 - 1.3205z + 0.6326)(z^2 - 1.0482z + 0.2959)}$$

The unit sample response corresponding to $H_d(z)$ is illustrated in Fig. 8.48. Use the Padé approximation method to approximate $H_d(z)$.

Solution We observe that the desired filter has $M = 4$ zeros and $N = 4$ poles. It is instructive to determine the coefficients in the Padé approximation when the number of zeros and/or poles are not identical to the desired number of filter parameters.

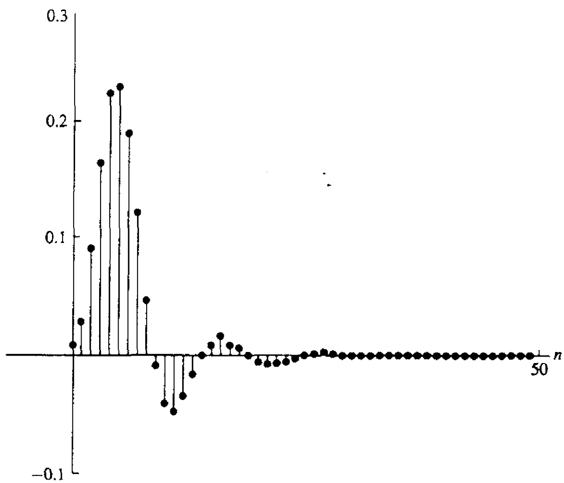

Figure 8.48 Impulse response $h_d(n)$ of digital Butterworth filter in Example 8.5.2.